The Wilson Loop Functions on Two Orientations of the Lattice

Abstract

The limits of the renormalised Wilson loop expectations constructed in the paper are covariant under the symmetries of the lattice, and nothing is known about the rotations which are not among them. The smallest instance in which the invariance under such a rotation is a statement about the limits is that of two parallel planar loops whose separation $(5, 0)$ is turned into $(3, 4)$. On a torus whose periods are tilted against the lattice, the reflection which exchanges the two configurations maps the lattice to a second lattice, and we show that the invariance is equivalent to the independence of the limit of the choice between these two lattices. If the blocking factor is a multiple of 5, the construction of the paper can be compared with a run of the renormalisation group which has the same lattices, cubes, and expansions and differs from it in one respect: its fine lattice is turned. We record, in the order of the proofs, the modifications which Bałaban's proofs require for this run. The present text contains the setting, the symmetries which the run retains, its averaging operations with their momentum representation, the description of its blocks as unions of cells with a Poincaré inequality, and, for the Gaussian theory without an external field on the whole lattice and on tori, the results of the paper on the scalar Green's functions and of the first paper on the propagators: the lower bounds, and the bounds on the propagators and their derivatives in L^2, in the supremum norm, and in Hölder norms, with exponential decay. One of Bałaban's methods, the passage to a box with Neumann conditions by reflections, is not available for a turned lattice. The last section lists the proofs which have not been examined.

1 Introduction

The paper constructs, for every ensemble $\mathcal{C}$ of admissible loops on the torus $(\mathbb{R}/T\mathbb{Z})^4$, the limit $W(\mathcal{C})$ of the renormalised expectations $W_K(\mathcal{C})$ as the lattice spacing $\varepsilon_K = uL^{-K}$ tends to zero (Paper, Theorem 1.5), and it shows that the limits are covariant under the translations by vectors of the lattice of the loops and under the reflections in coordinate hyperplanes (Paper, Corollary 1.6). One expects more. The lattice breaks the invariance of the continuum theory under rotations, and the limit in which the lattice is removed should restore it, as far as the torus permits. We ask whether this can be proved for the limits of the paper.

For most ensembles the question cannot be posed within the construction: a rotation which is not a symmetry of the lattice maps a path of the lattice to a path which is not one, and the limits are defined for paths of the lattice only. There is one family of ensembles for which it can. Let the loops lie in planes parallel to e_0 and e_1, and let the rotation turn the plane of e_2 and e_3 and fix the other two coordinates. It then maps each loop to a translate of itself, and it maps the ensemble to an ensemble of the lattice whenever it maps the positions of the planes to points of the lattice. This happens for rotations which are not symmetries of the lattice, because a circle around a point of the square lattice may pass through points which the symmetries of the lattice do not relate: $5^2 = 3^2 + 4^2$. The smallest instance is a

loop C with a parallel copy, and the question is whether

$$W(C, C + 5\varepsilon_q e_2) = W(C, C + 3\varepsilon_q e_2 + 4\varepsilon_q e_3) \,, \tag{1.1}$$

where ε_q is the spacing of the lattice of the loops. The two sides are two numbers of one lattice theory, and none of the properties which we know relates them. Nor could these properties relate them: the kernel $(d_2, d_3) \mapsto e^{-|d_2|-|d_3|}$ is continuous, it is invariant under the symmetries of the square lattice, and it is reflection positive in both coordinate directions, since $e^{-(x_2+y_2)}$ is a product and $e^{-|x_3-y_3|}$ is a positive definite kernel, and it takes different values at $(5, 0)$ and $(3, 4)$.

The first difficulty is the torus. The rotation which turns $(5, 0)$ into $(3, 4)$ is not an isometry of $(\mathbb{R}/T\mathbb{Z})^4$, and on this torus the two configurations are not congruent, so that there is no reason for (1.1) to hold there. We therefore change the torus. Writing $z = x_2 + ix_3$, the reflection $\rho\colon z \mapsto \frac{3+4i}{5}\bar{z}$ of the plane maps 5 to $3+4i$, and it is an isometry of the torus whose periods in this plane are $T(2 + i)$ and $iT(2 + i)$, a torus which the lattices of the paper fit as well. On this torus ρ maps the lattice Λ_K of spacing ε_K to a second lattice $\Lambda'_K = \rho\Lambda_K$, tilted against the first, and it maps the lattice theory on the one to the lattice theory on the other. The expectation of the configuration $(5, 0)$ on Λ_K is therefore equal to the expectation of the configuration $(3, 4)$ on Λ'_K, at every K and every coupling (Proposition 4.2). The configuration $(3, 4)$ consists of paths of both lattices, and (1.1) becomes the statement that the limit of its renormalised expectations is the same along the two families of lattices: the independence of the continuum limit of the orientation of the lattice, in the one case in which the two orientations share an observable.

A comparison of two lattice theories is what the paper does when it proves that the limit exists. It compares the construction on Λ_K with that on Λ_{K+1}, two runs of the renormalisation group which share every lattice but the finest (Paper, Section 9.1). The lattices Λ_K and Λ'_K share nothing, but if the blocking factor L is a multiple of 5, then Λ_K is a sublattice of Λ'_{K+1} (Lemma 2.2). A run which starts from the lattice theory on the tilted lattice Λ'_{K+1} and passes, by one renormalisation step, to the lattice Λ_K of the paper, and from there to $\Lambda_{K-1}, \Lambda_{K-2}, \ldots$, shares every lattice but the finest with the construction of the paper on Λ_K. Its coarse lattices, its cubes, its localisations, and its expansions are those of the paper. What differs is the fine lattice, on which every background field and every propagator of the run is defined: it is turned by ρ against everything else.

The proofs of Bałaban on which the paper rests are written for a fine lattice of which the coarser lattices are sublattices spanned by multiples of the same basis. Our aim is to record what changes in them when the fine lattice is turned, and only that: each statement which is used is assigned to one of three classes, those which hold with the same proof, those which hold after a modification, which we state and prove, and those which we have not examined. We proceed in the order of the proofs. Sections 2–4 contain the setting: the arithmetic of the two lattices, the torus, and the identity between the expectations which reduces (1.1) to a comparison of two runs. Section 5 determines the symmetries which the tilted run retains; they leave three invariant quadratic forms of the field strength where the symmetries of one lattice leave one, which affects the definition of the running coupling. Section 6 defines the averaging operations of the tilted run and proves the properties which replace those of [2, Sections A and B], and Section 7 gives their momentum representation, in which the turned lattice appears as one fixed trigonometric polynomial multiplying Bałaban's symbols and as a matrix in place of a scalar. Section 8 describes a block as a union of cells of 5^4 points, which gives contours inside the blocks and a Poincaré inequality, and Section 9 proves the L^2-bounds of [2] for the tilted run by counting paths. Section 10 goes through [2] formula by formula and Section 11 through the first two sections of [1]. Sections 12–14 then prove the

bounds on the scalar operator of the whole lattice which these two papers use, the kernel with its exponential decay, the equation which relates the scales, and the bounds in the supremum and Hölder norms; Section 15 bounds the kernel of the gauge-fixing projection, and Section 16 proves Bałaban's Proposition 1.2 on the propagator of the vector field. Section 18 lists what remains.

Throughout, G, L, u, m_T, $T = 2L^{m_T}u$, and $\varepsilon_K = uL^{-K}$ are as in Paper, Section 1.1; in particular L is odd.

2 The Two Lattices

We identify the plane of the coordinates x_2 and x_3 with $\mathbb{C}$ through $z = x_2 + ix_3$, and we write a point of $\mathbb{R}^4$ as (x_0, x_1, z). The points of this plane with integer coordinates are the Gaussian integers $\mathbb{Z}[i] = \mathbb{Z} + i\mathbb{Z}$. We put

$$\omega := \frac{3+4i}{5} = \frac{(2+i)^2}{5} = \frac{2+i}{2-i}\,, \qquad \rho(x_0, x_1, z) := (x_0, x_1, \omega\bar{z})\,. \tag{2.1}$$

Since $|\omega| = 1$ and $\omega\,\overline{\omega\bar{z}} = z$, the map ρ is a linear isometry of $\mathbb{R}^4$ with $\rho^2 = 1$. It fixes e_0 and e_1, it fixes $2 + i$ because $\omega(2 - i) = 2 + i$, and it maps $i(2 + i)$ to $\omega(-i)(2 - i) = -i(2 + i)$; it is therefore the reflection in the hyperplane spanned by e_0, e_1, and $2e_2 + e_3$. We write $e'_\nu := \rho e_\nu$, so that $e'_0 = e_0$, $e'_1 = e_1$, and, because $\omega\bar{1} = \omega$ and $\omega\bar{i} = (4 - 3i)/5$,

$$e'_2 = \tfrac{1}{5}(3e_2 + 4e_3)\,, \quad e'_3 = \tfrac{1}{5}(4e_2 - 3e_3)\,, \qquad e_2 = \tfrac{1}{5}(3e'_2 + 4e'_3)\,, \quad e_3 = \tfrac{1}{5}(4e'_2 - 3e'_3)\,, \tag{2.2}$$

the last two by applying ρ to the first two. In particular $\rho(5e_2) = 3e_2 + 4e_3$.

Lemma 2.1 (Gaussian Integers). *(i) For $w \in \mathbb{Z}[i]$, $\omega w \in \mathbb{Z}[i]$ if and only if $w \in (2-i)\mathbb{Z}[i]$. Consequently $\mathbb{Z}[i] \cap \omega\mathbb{Z}[i] = (2 + i)\mathbb{Z}[i]$.*

(ii) The group $(2 + i)\mathbb{Z}[i]$ has the index 5 in $\mathbb{Z}[i]$ and in $\omega\mathbb{Z}[i]$.

(iii) $\omega\,\overline{(2 + i)\mathbb{Z}[i]} = (2 + i)\mathbb{Z}[i]$.

(iv) $5\omega\mathbb{Z}[i] \subset \mathbb{Z}[i]$ and $5\mathbb{Z}[i] \subset \omega\mathbb{Z}[i]$, each with the index 25.

Proof. (i) Let $w = a + bi$ with integers a, b. Then $5\omega w = (3a - 4b) + (4a + 3b)i$, and $\omega w \in \mathbb{Z}[i]$ if and only if $3a \equiv 4b$ and $4a \equiv -3b$ modulo 5. Multiplying the first congruence by 2 and the second by 4, each of them is equivalent to $a \equiv 3b$, that is, to $a + 2b \equiv 0$ modulo 5. On the other hand $(2 - i)(m + ni) = (2m + n) + (2n - m)i$, for which $a + 2b = 5n$; and if $a + 2b = 5n$, then $m := 2n - b$ satisfies $2m + n = 5n - 2b = a$, so that $w = (2 - i)(m + ni)$. This is the first assertion. For such $w = (2 - i)w'$ we have $\omega w = (2 + i)w'$, so that the elements of $\omega\mathbb{Z}[i]$ which lie in $\mathbb{Z}[i]$ are those of $(2 + i)\mathbb{Z}[i]$.

(ii) The multiplication by $2 + i$ is the linear map of $\mathbb{Z}^2$ with the columns $(2, 1)$ and $(-1, 2)$, of determinant 5, so that its image has the index 5 in $\mathbb{Z}[i]$. The multiplication by ω^{-1} maps $\omega\mathbb{Z}[i]$ onto $\mathbb{Z}[i]$ and $(2 + i)\mathbb{Z}[i]$ onto $(2 - i)\mathbb{Z}[i]$, which has the index 5 in $\mathbb{Z}[i]$ for the same reason.

(iii) Complex conjugation maps $\mathbb{Z}[i]$ onto itself, so that $\omega\,\overline{(2 + i)\mathbb{Z}[i]} = \omega(2 - i)\mathbb{Z}[i] = (2 + i)\mathbb{Z}[i]$.

(iv) $5\omega = 3 + 4i \in \mathbb{Z}[i]$ gives the first inclusion, and the multiplication by $3 + 4i$ has the determinant 25. Since $\omega\bar{\omega} = 1$ and $5\bar{\omega} = 3 - 4i \in \mathbb{Z}[i]$, we have $5\mathbb{Z}[i] = \omega(5\bar{\omega})\mathbb{Z}[i] \subset \omega\mathbb{Z}[i]$, with the index of $(3 - 4i)\mathbb{Z}[i]$ in $\mathbb{Z}[i]$, which is 25. *q.e.d.*

For an integer $K \geq 0$ we consider the two lattices in $\mathbb{R}^4$

$$\Lambda_K \coloneqq \varepsilon_K \mathbb{Z}^4 = \varepsilon_K(\mathbb{Z}^2 \times \mathbb{Z}[i]) \,, \qquad \Lambda'_K \coloneqq \rho\Lambda_K = \varepsilon_K(\mathbb{Z}^2 \times \omega\mathbb{Z}[i]) \,. \tag{2.3}$$

The lattice Λ_K is the lattice of the paper before the periods are divided out, and Λ'_K is a lattice of the same spacing whose axes are the vectors e'_ν.

Lemma 2.2 (The Two Families of Lattices)**.** *(i) $\Lambda_K \cap \Lambda'_K = \varepsilon_K(\mathbb{Z}^2 \times (2+i)\mathbb{Z}[i])$, a sublattice of the index 5 in Λ_K and in Λ'_K, and ρ maps it onto itself.*

(ii) If L is a multiple of 5, then $\Lambda_K \subset \Lambda'_{K+1}$ and $\Lambda'_K \subset \Lambda_{K+1}$, each with the index L^4.

Proof. (i) is Lemma 2.1(i)–(iii). (ii) Since $\varepsilon_K = L\varepsilon_{K+1}$ and $L\mathbb{Z}[i] \subset 5\mathbb{Z}[i] \subset \omega\mathbb{Z}[i]$ by Lemma 2.1(iv), the definition (2.3) gives $\Lambda_K = \varepsilon_{K+1}(L\mathbb{Z}^2 \times L\mathbb{Z}[i]) \subset \Lambda'_{K+1}$. Both are lattices of rank four, and ρ preserves volumes, so that the index is the quotient $\varepsilon_K^4/\varepsilon_{K+1}^4 = L^4$ of the volumes of their cells. Applying ρ gives $\Lambda'_K \subset \Lambda_{K+1}$ with the same index. *q.e.d.*

Thus, for L a multiple of 5, the two families form one chain $\cdots \supset \Lambda'_{K+1} \supset \Lambda_K \supset \Lambda'_{K-1} \supset \Lambda_{K-2} \supset \ldots$, each lattice of the index L^4 in the one before, while no lattice of one family contains a lattice of the same index of the other.

3 One Torus for Both Lattices

The torus $(\mathbb{R}/T\mathbb{Z})^4$ of the paper is of no use for our question: $\rho(Te_2) = \frac{1}{5}T(3e_2 + 4e_3)$ is not a period, so that ρ does not act on it. We need a group of periods which ρ preserves and which lies in both lattices, and Lemma 2.2(i) provides one.

Definition 3.1 (The Tilted Torus)**.** Let

$$\Gamma \coloneqq T(\mathbb{Z}^2 \times (2+i)\mathbb{Z}[i]) \,, \qquad \mathbb{T}_\Gamma \coloneqq \mathbb{R}^4/\Gamma \,. \tag{3.1}$$

The periods of $\mathbb{T}_\Gamma$ are Te_0, Te_1, and the two vectors $T(2e_2 + e_3)$ and $T(-e_2 + 2e_3)$ of the plane, which are orthogonal and of the length $\sqrt{5}\,T$.

Lemma 3.2 (The Torus and the Lattices)**.** *(i) $\Gamma \subset \Lambda_K \cap \Lambda'_K$ for every $K \geq 0$, and the quotients Λ_K/Γ and Λ'_K/Γ have $5(T/\varepsilon_K)^4$ elements each.*

(ii) $\rho\Gamma = \Gamma$, so that ρ induces an isometry of $\mathbb{T}_\Gamma$, which maps Λ_K/Γ onto Λ'_K/Γ.

(iii) Γ has the index 5 in $T\mathbb{Z}^4$, so that $\mathbb{T}_\Gamma$ covers the torus of the paper five times.

Proof. (i) The number $T/\varepsilon_K = 2L^{m_T+K}$ is an integer, so that $\Gamma \subset \varepsilon_K(\mathbb{Z}^2 \times (2+i)\mathbb{Z}[i])$, which is $\Lambda_K \cap \Lambda'_K$ by Lemma 2.2(i). The index of Γ in this group is $(T/\varepsilon_K)^4$, and the index of this group in each of the two lattices is 5. (ii) is Lemma 2.1(iii), and $\rho\Lambda_K = \Lambda'_K$ by definition. (iii) is Lemma 2.1(ii). *q.e.d.*

4 The Identity between the Two Lattice Theories

We now define the lattice theory on $\mathbb{T}_\Gamma$ for each of the two lattices and show that ρ carries the one to the other. Let Ξ be Λ_K or Λ'_K. A *bond* of Ξ is an ordered pair $b = (x, y)$ of points of Ξ at the distance ε_K, with the reversed bond $b^{-1} = (y, x)$; the bonds from x are the eight

pairs $(x, x \pm \varepsilon_K e_\mu)$ for $\Xi = \Lambda_K$ and $(x, x \pm \varepsilon_K e'_\mu)$ for $\Xi = \Lambda'_K$. A *plaquette* of Ξ is a square of side ε_K with corners in Ξ, and ∂p is its boundary, a closed path of four bonds. The group $\Gamma \subset \Xi$ acts on the bonds and on the plaquettes by translations, and a translation by $\gamma \neq 0$ fixes no bond and no plaquette and reverses no bond. A *gauge field* on (Ξ, Γ) is a map U from the bonds of Ξ to G with $U(b^{-1}) = U(b)^{-1}$ and $U(b + \gamma) = U(b)$ for every $\gamma \in \Gamma$. The lattice theory at the coupling $g_0 > 0$ is the probability measure

$$\mathrm{d}\mu^{\Xi}_{\Gamma,g_0}(U) \coloneqq \frac{1}{Z^{\Xi}_{\Gamma,g_0}} \exp\Big(-\frac{1}{g_0^2} \sum_p (1 - \operatorname{Re}\operatorname{tr} U(\partial p))\Big)\, \mathrm{d}U \;, \tag{4.1}$$

where the sum runs over one plaquette from each orbit of Γ on the unoriented plaquettes of Ξ and $\mathrm{d}U$ is the product of the normalised Haar measures over one bond from each orbit of Γ on the unoriented bonds. Neither depends on the choices, because U is periodic, $\operatorname{Re}\operatorname{tr}$ is invariant under conjugation and, on unitary matrices, under inversion, and the Haar measure is invariant under inversion. This is the measure of Paper, (1.2), with the group of periods Γ in place of $T\mathbb{Z}^4$ and, for $\Xi = \Lambda'_K$, with the lattice turned by ρ. We write $\mathbb{E}^{\Xi}_{\Gamma,g_0}$ for its expectation. For a closed path C of bonds of Ξ in $\mathbb{R}^4$ and a character χ we put $W_\chi(C; U) \coloneqq \chi(U(C))$, where $U(C)$ is the ordered product of U over the bonds of C, and for a finite family $\mathcal{C} = ((C_1, \chi_1), \dots, (C_n, \chi_n))$ of such paths with characters, $W_{\mathcal{C}} \coloneqq \prod_i W_{\chi_i}(C_i)$. For a linear map O we write $O\mathcal{C}$ for the family of the images OC_i with the same characters.

Lemma 4.1 (Transport)**.** *Let Ξ be Λ_K or Λ'_K, and let O be a linear isometry of $\mathbb{R}^4$ with $O\Gamma = \Gamma$ for which $O\Xi$ is again one of the two lattices. Then, for every family $\mathcal{C}$ of closed paths of bonds of Ξ with characters and every $g_0 > 0$,*

$$\mathbb{E}^{O\Xi}_{\Gamma,g_0}[W_{O\mathcal{C}}] = \mathbb{E}^{\Xi}_{\Gamma,g_0}[W_{\mathcal{C}}] \;. \tag{4.2}$$

Proof. The map O is a bijection from the points of Ξ onto those of $O\Xi$ which preserves distances, so that it maps bonds to bonds, reversed bonds to reversed bonds, and plaquettes to plaquettes with their boundaries; and since $O(b+\gamma) = Ob + O\gamma$ with $O\Gamma = \Gamma$, it maps the orbits of Γ on the bonds and on the plaquettes of Ξ bijectively onto those of $O\Xi$. For a gauge field U on (Ξ, Γ) put $(O_*U)(b) \coloneqq U(O^{-1}b)$ for the bonds b of $O\Xi$. Then O_*U is a gauge field on $(O\Xi, \Gamma)$, the map O_* is a bijection between the two spaces of gauge fields which carries the product of the Haar measures of the one to that of the other, and $(O_*U)(O\gamma) = U(\gamma)$ for every path γ of bonds of Ξ. In particular $(O_*U)(\partial(Op)) = U(\partial p)$ for every plaquette p of Ξ, so that the sums in (4.1) for O_*U on $O\Xi$ and for U on Ξ agree, and O_* carries μ^{Ξ}_{Γ,g_0} to $\mu^{O\Xi}_{\Gamma,g_0}$. Since $W_{O\mathcal{C}}(O_*U) = W_{\mathcal{C}}(U)$, this gives (4.2). *q.e.d.*

We apply the lemma to $O = \rho$ and to the ensembles for which the question of the introduction can be posed. Let $q \geq 0$, and let $C_1, \dots, C_n$ be closed paths of bonds of Λ_q in the plane $\{z = 0\}$, so that all their bonds are parallel to e_0 or e_1. For $d = (d_1, \dots, d_n)$ with $d_j \in \varepsilon_q \mathbb{Z}[i]$ we write

$$\mathcal{C}_d \coloneqq ((C_1 + d_1, \chi_1), \dots, (C_n + d_n, \chi_n)) \;, \qquad \rho d \coloneqq (\omega \bar{d}_1, \dots, \omega \bar{d}_n) \;, \tag{4.3}$$

where $C_j + d_j$ is the translate of C_j to the plane $\{z = d_j\}$. Since ρ fixes the plane $\{z = 0\}$ pointwise and is linear, $\rho(C_j + d_j) = C_j + \omega \bar{d}_j$, that is, $\rho \mathcal{C}_d = \mathcal{C}_{\rho d}$.

Proposition 4.2 (Reduction)**.** *Let $q \geq 0$, let $C_1, \dots, C_n$ and $\mathcal{C}_d$ be as above, and let $d_j \in \varepsilon_q(2+i)\mathbb{Z}[i]$ for every j. Then $\rho d_j \in \varepsilon_q(2+i)\mathbb{Z}[i]$ for every j, the families $\mathcal{C}_d$ and $\mathcal{C}_{\rho d}$ consist of closed paths of bonds of both Λ_K and Λ'_K for every $K \geq q$, and for every $K \geq q$ and every $g_0 > 0$,*

$$\mathbb{E}^{\Lambda_K}_{\Gamma,g_0}[W_{\mathcal{C}_d}] = \mathbb{E}^{\Lambda'_K}_{\Gamma,g_0}[W_{\mathcal{C}_{\rho d}}] \;. \tag{4.4}$$

Consequently

$$\mathbb{E}^{\Lambda_K}_{\Gamma,g_0}[W_{\mathcal{C}_{\rho d}}] - \mathbb{E}^{\Lambda_K}_{\Gamma,g_0}[W_{\mathcal{C}_d}] = \mathbb{E}^{\Lambda_K}_{\Gamma,g_0}[W_{\mathcal{C}_{\rho d}}] - \mathbb{E}^{\Lambda'_K}_{\Gamma,g_0}[W_{\mathcal{C}_{\rho d}}] \,. \tag{4.5}$$

Proof. By Lemma 2.1(iii), $\rho d_j = \omega \bar{d}_j \in \varepsilon_q(2+i)\mathbb{Z}[i]$. The sites of $C_j + d_j$ are the points (x_0, x_1, d_j) with $(x_0, x_1, 0)$ a site of C_j; they lie in $\varepsilon_q(\mathbb{Z}^2 \times (2+i)\mathbb{Z}[i]) = \Lambda_q \cap \Lambda'_q$, and the bonds of $C_j + d_j$ are parallel to $e_0 = e'_0$ or $e_1 = e'_1$ and of the length ε_q. A straight path of the length ε_q between two points of Λ_q, or of Λ'_q, along one of its axes is a path of L^{K-q} bonds of Λ_K, or of Λ'_K; so $C_j + d_j$ is a closed path of bonds of both lattices, and the same holds for $C_j + \rho d_j$. Lemma 4.1 with $\Xi = \Lambda_K$ and $O = \rho$, for which $O\Gamma = \Gamma$ by Lemma 3.2(ii) and $O\Xi = \Lambda'_K$, gives (4.4) because $\rho\mathcal{C}_d = \mathcal{C}_{\rho d}$, and (4.5) follows by subtracting both sides of (4.4) from $\mathbb{E}^{\Lambda_K}_{\Gamma,g_0}[W_{\mathcal{C}_{\rho d}}]$. *q.e.d.*

The left side of (4.5) measures how far the lattice theory on Λ_K is from being invariant under ρ on this ensemble, and the right side how much the expectation of one ensemble differs between the two lattices. The first instance in which $\mathcal{C}_{\rho d}$ is not the image of $\mathcal{C}_d$ under a symmetry of Λ_q is $n = 2$, $d_1 = 0$, and $d_2 = 5\varepsilon_q = (2+i)(2-i)\varepsilon_q$, for which $\rho d_2 = (3+4i)\varepsilon_q$; this is (1.1). The step factors of the paper do not depend on the position of a loop (Paper, Proposition 7.13), so that $C_j + d_j$ and $C_j + \rho d_j$ have the same normalisation on Λ_K, and (1.1) is the statement that the left side of (4.5), divided by this normalisation, tends to zero as $K \to \infty$ along the couplings of the paper. On the right side, the first term is the object of the construction of the paper on Λ_K, carried out on the torus $\mathbb{T}_\Gamma$, and the second, read at $K+1$, is the object of the tilted run of the introduction.

5 The Symmetries of the Tilted Run

The tilted run has only those symmetries which preserve the lattices of both families. We determine them, and what they leave of the argument by which a single coupling describes the leading term of an effective action.

Lemma 5.1 (The Common Symmetries)**.** *Let $\mathfrak{G}_0$ be the group of the linear isometries O of $\mathbb{R}^4$ with $O\mathbb{Z}^4 = \mathbb{Z}^4$ and $O\rho\mathbb{Z}^4 = \rho\mathbb{Z}^4$. Its elements are the maps*

$$(x_0, x_1, z) \mapsto (A(x_0, x_1), i^k z) \,, \tag{5.1}$$

with A one of the eight signed permutation matrices of the first two coordinates and $k \in \{0,1,2,3\}$. It has 32 elements, it contains the reflections in the hyperplanes $\{x_0 = 0\}$ and $\{x_1 = 0\}$, and it contains no reflection of the plane of x_2 and x_3.

Proof. An isometry O with $O\mathbb{Z}^4 = \mathbb{Z}^4$ permutes the eight vectors $\pm e_\mu$, the elements of $\mathbb{Z}^4$ of the length 1, and an isometry with $O\rho\mathbb{Z}^4 = \rho\mathbb{Z}^4$ permutes the eight vectors $\pm e'_\nu$. By (2.2) the only vectors common to the two sets are $\pm e_0$ and $\pm e_1$. An element O of $\mathfrak{G}_0$ permutes both sets, hence their intersection, so that $Oe_0, Oe_1 \in \{\pm e_0, \pm e_1\}$; the map O therefore preserves the plane of e_0 and e_1 and its orthogonal complement, and it acts on the first as a signed permutation matrix A and on the second as a signed permutation matrix of e_2 and e_3, that is, as $z \mapsto i^k z$ or as $z \mapsto i^k \bar{z}$. Every such map preserves $\mathbb{Z}^4$, and it preserves $\rho\mathbb{Z}^4 = \mathbb{Z}^2 \times \omega\mathbb{Z}[i]$ if and only if its second part preserves $\omega\mathbb{Z}[i]$. The maps $z \mapsto i^k z$ do. The maps $z \mapsto i^k \bar{z}$ send $\omega\mathbb{Z}[i]$ to $\bar{\omega}\mathbb{Z}[i]$, which is equal to $\omega\mathbb{Z}[i]$ only if $\omega^2 \in \mathbb{Z}[i]$, and $\omega^2 = (-7+24i)/25$. This gives (5.1) and the count $8 \cdot 4$. The reflections named are the cases $A = \mathrm{diag}(-1, 1)$ and $A = \mathrm{diag}(1, -1)$ with $k = 0$. *q.e.d.*

The group of all symmetries of $\mathbb{Z}^4$, the signed permutation matrices, has 384 elements and exchanges any two coordinate planes. The group $\mathfrak{G}_0$ does not, and this has the following consequence. Let $F = (F_{\mu\nu})$ be an antisymmetric tensor, on which an isometry O acts by $(OF)_{\mu\nu} = \sum_{\kappa,\lambda} O_{\mu\kappa} O_{\nu\lambda} F_{\kappa\lambda}$.

Lemma 5.2 (Invariant Quadratic Forms)**.** *The real quadratic forms of the antisymmetric tensors F which are invariant under $\mathfrak{G}_0$ are the combinations*

$$\alpha F_{01}^2 + \beta F_{23}^2 + \lambda \sum_{a\in\{0,1\}} \sum_{j\in\{2,3\}} F_{aj}^2 \tag{5.2}$$

with real α, β, λ, a space of dimension three. Those which are invariant under all signed permutation matrices are the multiples of $\sum_{\mu<\nu} F_{\mu\nu}^2$.

Proof. Let q be an invariant quadratic form, a combination of the products of two of the six components F_{01}, F_{23}, F_{aj}. The three elements θ_0, θ_1, and σ of $\mathfrak{G}_0$ which change the sign of x_0, of x_1, and of z multiply each component by a sign: F_{01} by $(-,-,+)$, F_{23} by $(+,+,+)$, F_{0j} by $(-,+,-)$, and F_{1j} by $(+,-,-)$. A product of two components is invariant under the three only if the two have the same signs, so that

$$\mathsf{q}(F) = \alpha F_{01}^2 + \beta F_{23}^2 + \sum_{j,j'} (m^0_{jj'} F_{0j} F_{0j'} + m^1_{jj'} F_{1j} F_{1j'}) \tag{5.3}$$

with symmetric matrices m^0 and m^1. The element of $\mathfrak{G}_0$ which exchanges x_0 and x_1 exchanges F_{0j} and F_{1j}, so that $m^0 = m^1 =: m$. The element $z \mapsto iz$ maps e_2 to e_3 and e_3 to $-e_2$, so that it replaces the vector (F_{a2}, F_{a3}) by $(-F_{a3}, F_{a2})$; the invariance of (5.3) then requires $m_{22} = m_{33}$ and $m_{23} = -m_{23}$, that is, $m = \lambda \mathbf{1}$. This is (5.2). Conversely each of the three forms is invariant: an element (5.1) multiplies F_{01} by $\det A = \pm 1$, fixes F_{23}, because $z \mapsto i^k z$ has the determinant 1, and replaces the matrix (F_{aj}) by its product with A on one side and with a rotation matrix on the other, which preserves the sum of the squares of its entries. Finally a signed permutation matrix which exchanges the planes of two pairs of coordinates exchanges the corresponding components up to signs, so that the invariance under all of them forces $\alpha = \beta = \lambda$. *q.e.d.*

Modification of [8, (1.20)–(1.22), p. 264]. Bałaban defines the coefficient β_{j+1} of the flow of the coupling as a second derivative of the Fourier transform of the vacuum polarisation $\Pi_{j+1,\mu\nu}$ of a step, for one pair $\mu \neq \nu$, and the definition does not depend on the pair because, by his (1.21), the vacuum polarisation is covariant under the rotations which leave the lattice invariant. In the tilted run the covariance holds for the elements of $\mathfrak{G}_0$ only, and by Lemma 5.2 the second derivatives define three numbers, one for each of the forms (5.2): those of the pair $(0,1)$, of the pair $(2,3)$, and of the four pairs (a,j). We do not know whether the three agree, and the consequences for the inductive description of the densities have not been examined.

6 The Averages of the Tilted Run

From here on L is a multiple of 5. After $k \geq 1$ steps of the tilted run, and in the unit of the lattice which it has reached, the situation is that of [2, Section B] with one change: the unit lattice is $\mathbb{Z}^4$ and the lattices of the intermediate steps are $L^{-j}\mathbb{Z}^4$, $1 \leq j \leq k-1$, as there, but the fine lattice is

$$T_\eta \coloneqq \eta\rho\mathbb{Z}^4 = \eta(\mathbb{Z}^2 \times \omega\mathbb{Z}[i]) \,, \qquad \eta = L^{-k} \,, \tag{6.1}$$

in place of $\eta\mathbb{Z}^4$. Its bonds are the pairs $\langle x, x \pm \eta e'_\nu \rangle$. We call a number a a *scale* if $a/(5\eta)$ and $1/a$ are integers; the numbers L^{-j}, $0 \le j \le k-1$, are scales, and so is 5η. For a scale a we have $a\mathbb{Z}^4 \subset T_\eta$, because a/η is a multiple of 5 and $5\mathbb{Z}[i] \subset \omega\mathbb{Z}[i]$ by Lemma 2.1(iv). A real field is a function A on the bonds of T_η with $A(b^{-1}) = -A(b)$, and $A(\gamma) \coloneqq \sum_{b\in\gamma} A(b)$ for a path γ of bonds. We consider infinite lattices in this section and the next; all the operations commute with the translations by $\mathbb{Z}^4$ and therefore act on the fields which are periodic with respect to a subgroup of $\mathbb{Z}^4$.

Bałaban's averages are sums, over the points x of a cube of the fine lattice, of the values of a function at x or of the sums of a field over the straight segment $[x, x(b)]$ from x to its translate by a bond b of the coarse lattice [2, (1.6), (1.11), (1.18), pp. 18–20]. In the tilted run the cubes keep their meaning, as the sets of the points of T_η in a cube of the coarse lattice, but for the directions e_2 and e_3 the segment does not consist of bonds of T_η, and we replace it by a path which follows it.

Definition 6.1 (Blocks, Staircases, and Averages)**.** Let a be a scale.

(i) The *block* of $y \in a\mathbb{Z}^4$ is $B_a(y) \coloneqq \{x \in T_\eta \,|\, y_\mu \le x_\mu < y_\mu + a,\ \mu = 0, \dots, 3\}$.

(ii) For $x \in T_\eta$ the *staircases* from x are the following paths of bonds of T_η. For $\mu \in \{0, 1\}$, $\sigma^a_\mu(x)$ is the straight path of a/η bonds in the direction e_μ. The path $\sigma^a_2(x)$ repeats $a/(5\eta)$ times the seven steps $\eta e'_3, \eta e'_2, \eta e'_3, \eta e'_2, \eta e'_3, \eta e'_2, \eta e'_3$, and $\sigma^a_3(x)$ repeats $a/(5\eta)$ times the seven steps $\eta e'_2, -\eta e'_3, \eta e'_2, -\eta e'_3, \eta e'_2, -\eta e'_3, \eta e'_2$.

(iii) For a function λ on T_η and $y \in a\mathbb{Z}^4$,

$$(Q'_a\lambda)(y) \coloneqq \Big(\frac{\eta}{a}\Big)^4 \sum_{x\in B_a(y)} \lambda(x)\ . \tag{6.2}$$

(iv) For a real field A and a bond $b = \langle y, y + ae_\mu \rangle$ of $a\mathbb{Z}^4$,

$$(Q_a A)(b) \coloneqq \Big(\frac{\eta}{a}\Big)^5 \sum_{x\in B_a(y)} A(\sigma^a_\mu(x))\ , \tag{6.3}$$

and $(Q_a A)(b^{-1}) \coloneqq -(Q_a A)(b)$.

We write $Q'_k \coloneqq Q'_1$ and $Q_k \coloneqq Q_1$.

For $a = 1$ these are the operators Q'_k of [2, p. 20] and Q_k of [2, (1.18), p. 20], with T_η turned and with $\sigma^1_\mu(x)$ in place of $[x, x(b)]$, and for the directions e_0 and e_1 nothing has changed. For two scales a and a' with a'/a an integer we also need Bałaban's own averages between the lattices $a\mathbb{Z}^4$ and $a'\mathbb{Z}^4$, which we denote by

$$(Q'^{\mathrm{st}}_{a,a'}\lambda)(y) \coloneqq \Big(\frac{a}{a'}\Big)^4 \sum_z \lambda(z)\ , \qquad (Q^{\mathrm{st}}_{a,a'}B)(\langle y, y+a'e_\mu\rangle) \coloneqq \Big(\frac{a}{a'}\Big)^5 \sum_z B([z, z+a'e_\mu])\ , \tag{6.4}$$

the sums over the points $z \in a\mathbb{Z}^4$ with $y_\mu \le z_\mu < y_\mu + a'$ for every μ, for functions λ and fields B on $a\mathbb{Z}^4$.

Lemma 6.2 (Blocks and Staircases)**.** *Let a and a' be scales with a'/a an integer.*

(i) The blocks $B_a(y)$, $y \in a\mathbb{Z}^4$, are pairwise disjoint, their union is T_η, each has $(a/\eta)^4$ points, and $B_a(y + w) = B_a(y) + w$ for $w \in a\mathbb{Z}^4$. The block $B_{a'}(y)$ is the union of the blocks $B_a(z)$ over the points $z \in a\mathbb{Z}^4$ with $y_\mu \le z_\mu < y_\mu + a'$.

(ii) The staircase $\sigma_\mu^a(x)$ ends at $x + ae_\mu$, and $\sigma_\mu^{a'}(x)$ is the concatenation of the staircases $\sigma_\mu^a(x + nae_\mu)$, $0 \le n < a'/a$. Each site of $\sigma_\mu^a(x)$ is at a distance of at most $\frac{3}{5}\eta$ from the segment $[x, x + ae_\mu]$.

Proof. (i) Since $a\mathbb{Z}^4$ is a subgroup of T_η and the cube $y + [0, a)^4$ contains exactly one point of every class of $\mathbb{R}^4$ modulo $a\mathbb{Z}^4$, the block $B_a(y)$ contains exactly one point of every class of T_η modulo $a\mathbb{Z}^4$. The number of these classes is the quotient a^4/η^4 of the volumes of the cells of the two lattices, because ρ preserves volumes. The remaining assertions follow from the corresponding ones for the cubes.

(ii) For $\mu \in \{0, 1\}$ there is nothing to prove. By (2.2), seven steps of σ_2^a add up to $\eta(3e_2' + 4e_3') = 5\eta e_2$ and seven steps of σ_3^a to $\eta(4e_2' - 3e_3') = 5\eta e_3$, which gives the end point and the concatenation. The distance of a point $\eta(ne_2' + n'e_3')$ from the line through the origin in the direction $3e_2' + 4e_3'$ is $\eta|4n - 3n'|/5$. After one to seven steps of σ_2^a from the origin the pair (n, n') is $(0, 1)$, $(1, 1)$, $(1, 2)$, $(2, 2)$, $(2, 3)$, $(3, 3)$, $(3, 4)$, with $|4n-3n'| = 3, 1, 2, 2, 1, 3, 0$, and the foot of the perpendicular lies between the origin and $5\eta e_2$ in each case; the eighth site lies on the line, so that the same holds in every period. For σ_3^a the pairs are $(1, 0)$, $(1, -1)$, $(2, -1)$, $(2, -2)$, $(3, -2)$, $(3, -3)$, $(4, -3)$, and the distance from the line in the direction $4e_2' - 3e_3'$ is $\eta|3n + 4n'|/5$, with the same values. *q.e.d.*

The next lemma replaces the calculation [2, (1.16)–(1.18), p. 20] by which a composition of averages is an average, and it contains a second statement which has no counterpart there: the tilted average is one of Bałaban's averages, preceded by an average over cells of 5^4 points which is the same at every step.

Lemma 6.3 (Composition)**.** *For scales a and a' with a'/a an integer,*

$$Q_{a'} = Q^{\mathrm{st}}_{a,a'} Q_a \ , \qquad Q'_{a'} = Q'^{\mathrm{st}}_{a,a'} Q'_a \ . \tag{6.5}$$

In particular $Q_k = Q^{\mathrm{st}}_{L^{-1},1} Q_{L^{-1}}$ and $Q_k = Q^{\mathrm{st}}_{5\eta,1} Q_{5\eta}$, and likewise for Q'_k.

Proof. Let $b = \langle y, y + a'e_\mu \rangle$, and let z run over the points of $a\mathbb{Z}^4$ with $y_\nu \le z_\nu < y_\nu + a'$. The segment $[z, z + a'e_\mu]$ consists of the a'/a bonds $\langle z + nae_\mu, z + (n+1)ae_\mu \rangle$ of $a\mathbb{Z}^4$, so that by (6.4) and (6.3)

$$\begin{aligned} (Q^{\mathrm{st}}_{a,a'} Q_a A)(b) &= \Big(\frac{a}{a'}\Big)^5 \Big(\frac{\eta}{a}\Big)^5 \sum_z \sum_{n=0}^{a'/a-1} \sum_{x \in B_a(z+nae_\mu)} A(\sigma_\mu^a(x)) \\ &= \Big(\frac{\eta}{a'}\Big)^5 \sum_z \sum_{x \in B_a(z)} \sum_{n=0}^{a'/a-1} A(\sigma_\mu^a(x + nae_\mu)) \ , \end{aligned} \tag{6.6}$$

where we have used $B_a(z + nae_\mu) = B_a(z) + nae_\mu$. By Lemma 6.2(ii) the inner sum is $A(\sigma_\mu^{a'}(x))$, and by Lemma 6.2(i) the points x run over $B_{a'}(y)$; this is $(Q_{a'}A)(b)$. The proof for Q' is the last assertion of Lemma 6.2(i). *q.e.d.*

Lemma 6.4 (The Identities of the Averages)**.** *Let a be a scale, and write $(\partial^\eta \lambda)(b) := \eta^{-1}(\lambda(b_+) - \lambda(b_-))$ for a function λ on T_η and a bond $b = \langle b_-, b_+ \rangle$ of T_η, and ∂^a for the same operation on $a\mathbb{Z}^4$.*

(i) $Q_a \partial^\eta = \partial^a Q'_a$.

(ii) Q'_a *maps the constant function* 1 *to* 1*, and so does its adjoint. For* $v \in \mathbb{R}^4$*,* Q_a *maps the constant field* $A(\langle x, x+\eta e'_\nu\rangle) \coloneqq v \cdot e'_\nu$ *on* T_η *to the constant field* $B(\langle y, y+ae_\mu\rangle) \coloneqq v \cdot e_\mu$ *on* $a\mathbb{Z}^4$*, and its adjoint maps* B *to* A*. Here the adjoints are taken with respect to the sums over the points, or the positive bonds, with the weights* η^4 *on* T_η *and* a^4 *on* $a\mathbb{Z}^4$*.*

(iii) Q_a *and* Q'_a *commute with the translations by the vectors of* $a\mathbb{Z}^4$*.*

Proof. (i) The sum of $\partial^\eta\lambda$ over a path from x to x' is $\eta^{-1}(\lambda(x')-\lambda(x))$, so that $(Q_a\partial^\eta\lambda)(\langle y, y+ae_\mu\rangle) = (\eta/a)^5\eta^{-1}\sum_{x\in B_a(y)}(\lambda(x+ae_\mu)-\lambda(x))$ by Lemma 6.2(ii). Since $x \mapsto x+ae_\mu$ maps $B_a(y)$ onto $B_a(y+ae_\mu)$, this is $a^{-1}((Q'_a\lambda)(y+ae_\mu)-(Q'_a\lambda)(y))$.

(ii) $Q'_a 1 = 1$ is the count of Lemma 6.2(i), and $(Q'^*_a\omega)(x) = \omega(y)$ for $x \in B_a(y)$, because every point lies in one block. The sum of the constant field A over a path from x to x' is $\eta^{-1}v\cdot(x'-x)$, so that every term of (6.3) is $\eta^{-1}av\cdot e_\mu$, and $(\eta/a)^5(a/\eta)^4\eta^{-1}a = 1$. For the adjoint we count the staircases through a bond. By (6.3) and the weights, $(Q^*_aB)(b') = (\eta/a)\sum_\mu\sum_x \pm B(\langle y(x), y(x)+ae_\mu\rangle)$ for a positive bond b' of T_η, the inner sum over the points $x \in T_\eta$ whose staircase $\sigma^a_\mu(x)$ contains b' or its reverse, with the sign $-$ in the second case, and with $y(x)$ the corner of the block of x. A bond in the direction e_0 or e_1 lies on a/η staircases of that direction and on no other. Each period of σ^a_2 has three steps $\eta e'_2$ and four steps $\eta e'_3$, and each period of σ^a_3 has four steps $\eta e'_2$ and three steps $-\eta e'_3$; since every translate of a staircase by a vector of T_η is a staircase, a bond in the direction e'_2 lies on $3a/(5\eta)$ staircases σ^a_2 and on $4a/(5\eta)$ staircases σ^a_3, and a bond in the direction e'_3 on $4a/(5\eta)$ staircases σ^a_2 and, reversed, on $3a/(5\eta)$ staircases σ^a_3. For the constant field B this gives the values $\frac{1}{5}(3v\cdot e_2 + 4v\cdot e_3) = v\cdot e'_2$ and $\frac{1}{5}(4v\cdot e_2 - 3v\cdot e_3) = v\cdot e'_3$ by (2.2).

(iii) The blocks and the staircases are carried to each other by these translations. *q.e.d.*

Statement (i) is [2, (1.55), p. 27], and with it [2, (1.9), (1.13), (1.20)]. Statement (ii) says that the average of a constant field is the same constant vector, with its components in the basis e'_ν replaced by those in the basis e_μ. On the constant fields Q_a is therefore an isometry with $Q^*_aQ_a = 1$, where Bałaban's average is the identity, and Q_a and Q'_a map the orthogonal complements of the constants to the orthogonal complements of the constants.

7 The Momentum Representation

Bałaban's bounds on the operators built from Q_k, Q'_k, and the Laplace operator of the fine lattice are read off from their momentum representation, in which Q'_k is the multiplication by the function u_k of [2, (1.31), p. 23] and Q_k the multiplication of the component μ by u_kv_μ [2, (1.61), p. 28]. Both are products of quotients of sines, and their properties are evident from this form. For the tilted averages the factorisation $Q_k = Q^{\mathrm{st}}_{5\eta,1}Q_{5\eta}$ of Lemma 6.3 gives the corresponding form: Bałaban's functions for the lattice of the spacing 5η, multiplied by one trigonometric polynomial of 25 terms for the cell, and with a matrix of polynomials of at most four terms in place of the scalar v_μ in the two turned directions.

For $p \in \mathbb{R}^4$ we write $e_p(x) \coloneqq e^{ip\cdot x}$, and for a number $a > 0$

$$\begin{aligned} \partial^a_\mu(p) &\coloneqq \frac{e^{iap_\mu}-1}{a}\,, \qquad & \partial'_\nu(p) &\coloneqq \frac{e^{i\eta p\cdot e'_\nu}-1}{\eta}\,, \\ \Delta(p) &\coloneqq \sum_\nu |\partial'_\nu(p)|^2\,, \qquad & \Delta_0(p) &\coloneqq \sum_\mu |\partial^1_\mu(p)|^2\,. \end{aligned} \tag{7.1}$$

The functions ∂'_ν and Δ are the symbols of the difference operators and of the Laplace operator of T_η, and ∂^1_μ and Δ_0 those of the unit lattice, as in [2, (1.31), p. 23, and p. 24]. For numbers a and a' with a'/a an integer we put

$$D^{a,a'}_\mu(p) := \frac{a}{a'} \sum_{n=0}^{a'/a-1} e^{ianp_\mu} , \tag{7.2}$$

which is $\partial^{a'}_\mu(p)/\partial^a_\mu(p)$ wherever $\partial^a_\mu(p) \neq 0$ and satisfies $D^{a,a''}_\mu = D^{a,a'}_\mu D^{a',a''}_\mu$. Bałaban's functions are $u_k = \prod_\mu D^{\eta,1}_\mu$ and $v_\mu = D^{\eta,1}_\mu$. The cell enters through the set $\mathsf{C} := \{c \in \omega\mathbb{Z}[i] \,|\, 0 \leq \operatorname{Re} c < 5,\ 0 \leq \operatorname{Im} c < 5\}$, which has 25 elements by Lemma 6.2(i) for $a = 5\eta$, and through the functions of $\theta = (\theta_2, \theta_3)$

$$\begin{aligned}
\kappa(\theta) &:= \frac{1}{25} \sum_{c\in\mathsf{C}} e^{i\theta\cdot c} , \\
s_{22}(\theta) &:= e^{i\theta\cdot e'_3} + e^{i\theta\cdot(e'_2+2e'_3)} + e^{i\theta\cdot(2e'_2+3e'_3)} , \\
s_{23}(\theta) &:= 1 + e^{i\theta\cdot(e'_2+e'_3)} + e^{i\theta\cdot(2e'_2+2e'_3)} + e^{i\theta\cdot(3e'_2+3e'_3)} , \\
s_{32}(\theta) &:= 1 + e^{i\theta\cdot(e'_2-e'_3)} + e^{i\theta\cdot(2e'_2-2e'_3)} + e^{i\theta\cdot(3e'_2-3e'_3)} , \\
s_{33}(\theta) &:= -e^{i\theta\cdot(e'_2-e'_3)} - e^{i\theta\cdot(2e'_2-2e'_3)} - e^{i\theta\cdot(3e'_2-3e'_3)} ,
\end{aligned} \tag{7.3}$$

where c is read as the vector $(\operatorname{Re} c, \operatorname{Im} c)$ and $\theta \cdot e'_\nu$ is the product with the components of e'_ν in the plane.

Lemma 7.1 (The Symbols of the Tilted Averages)**.** *Let $p \in \mathbb{R}^4$.*

(i) $Q'_k e_p = u_k(p)\, e_p$ on $\mathbb{Z}^4$, with

$$\begin{aligned}
u_k(p) &:= \sum_{x\in B_1(0)} \eta^4 e^{ip\cdot x} \\
&= D^{\eta,1}_0(p)\, D^{\eta,1}_1(p)\, D^{5\eta,1}_2(p)\, D^{5\eta,1}_3(p)\, \kappa(\eta p_2, \eta p_3) .
\end{aligned} \tag{7.4}$$

(ii) For $w = (w_\nu) \in \mathbb{C}^4$ and the field $A(\langle x, x+\eta e'_\nu\rangle) := w_\nu e^{ip\cdot x}$, we have $(Q_k A)(\langle y, y+e_\mu\rangle) = u_k(p) \sum_\nu V_{\mu\nu}(p) w_\nu\, e^{ip\cdot y}$, where $V_{\mu\mu}(p) := D^{\eta,1}_\mu(p)$ for $\mu \in \{0,1\}$,

$$V_{\mu\nu}(p) := \tfrac{1}{5} D^{5\eta,1}_\mu(p)\, s_{\mu\nu}(\eta p_2, \eta p_3) \tag{7.5}$$

for $\mu, \nu \in \{2,3\}$, and $V_{\mu\nu} := 0$ for the other pairs.

(iii) $\sum_\nu V_{\mu\nu}(p)\, \partial'_\nu(p) = \partial^1_\mu(p)$ for every μ.

(iv) u_k and V are periodic with respect to $2\pi\eta^{-1}\rho\mathbb{Z}^4$, which is a subgroup of $2\pi\mathbb{Z}^4$; $u_k(0) = 1$; $u_k(l) = 0$ for every $l \in 2\pi\mathbb{Z}^4$ which does not lie in $2\pi\eta^{-1}\rho\mathbb{Z}^4$; and $V_{\mu\nu}(0) = e_\mu \cdot e'_\nu$.

Proof. (i) By (6.2), $(Q'_k e_p)(y) = \sum_{x\in B_1(0)} \eta^4 e^{ip\cdot(y+x)}$, which is the first equality. By Lemma 6.2(i), $B_1(0)$ is the disjoint union of the sets $z + B_{5\eta}(0)$ over the $z \in 5\eta\mathbb{Z}^4$ with $0 \leq z_\mu < 1$, and $B_{5\eta}(0)$ is the set of the points $\eta(n_0, n_1, c)$ with $n_0, n_1 \in \{0, \dots, 4\}$ and $c \in \mathsf{C}$. Hence

$$\begin{aligned}
u_k(p) &= \Big(\sum_z (5\eta)^4 e^{ip\cdot z}\Big)\Big(\frac{1}{5^4} \sum_{n_0,n_1=0}^{4} \sum_{c\in\mathsf{C}} e^{i\eta(n_0 p_0 + n_1 p_1)} e^{i\eta(p_2,p_3)\cdot c}\Big) \\
&= \prod_{\mu=0}^{3} D^{5\eta,1}_\mu(p) \cdot D^{\eta,5\eta}_0(p)\, D^{\eta,5\eta}_1(p)\, \kappa(\eta p_2, \eta p_3) ,
\end{aligned} \tag{7.6}$$

and $D_\mu^{\eta,5\eta} D_\mu^{5\eta,1} = D_\mu^{\eta,1}$.

(ii) For $\mu \in \{0,1\}$ the sum of A over $\sigma_\mu^1(x)$ is $w_\mu e^{ip\cdot x} \sum_{n<1/\eta} e^{i\eta n p_\mu} = \eta^{-1} D_\mu^{\eta,1}(p) w_\mu e^{ip\cdot x}$, and (6.3) with (i) gives the assertion. For $\mu = 2$, the bonds of the first period of $\sigma_2^1(x)$ start at the points $x+\eta c$ with $c = 0$, e_3', $e_2'+e_3'$, $e_2'+2e_3'$, $2e_2'+2e_3'$, $2e_2'+3e_3'$, $3e_2'+3e_3'$, and they have the directions $e_3', e_2', e_3', e_2', e_3', e_2', e_3'$; the sum of A over them is $e^{ip\cdot x}(s_{22}(\eta p)w_2 + s_{23}(\eta p)w_3)$. The n-th period is the translate of the first by $5\eta n e_2$, so that the sum of A over $\sigma_2^1(x)$ is this number multiplied by $\sum_{n<1/(5\eta)} e^{5i\eta n p_2} = (5\eta)^{-1} D_2^{5\eta,1}(p)$. For $\mu = 3$ the steps in the direction e_2' start at the points $x + \eta j(e_2' - e_3')$, $j = 0, \dots, 3$, and a step $-\eta e_3'$ from $x+\eta c$ is the reversed bond $\langle x+\eta(c-e_3'), x+\eta c\rangle$, on which A takes the value $-w_3 e^{ip\cdot(x+\eta(c-e_3'))}$; the three such steps end at $x + \eta j(e_2' - e_3')$, $j = 1, 2, 3$. This gives s_{32} and s_{33}.

(iii) Multiply each term of $s_{\mu\nu}(\theta)$ by $e^{i\theta\cdot e_\nu'} - 1$ and add: each step of a period from c to c' contributes $e^{i\theta\cdot c'} - e^{i\theta\cdot c}$, a step $-\eta e_3'$ included, since $-(e^{i\theta\cdot e_3'} - 1)e^{i\theta\cdot(c-e_3')} = e^{i\theta\cdot(c-e_3')} - e^{i\theta\cdot c}$. The sum over a period telescopes to $e^{5i\theta_\mu} - 1$, because the period ends at $5e_\mu$. Hence $\sum_\nu V_{\mu\nu}\partial_\nu' = \frac{1}{5} D_\mu^{5\eta,1}(p)\eta^{-1}(e^{5i\eta p_\mu} - 1) = D_\mu^{5\eta,1}(p)\, \partial_\mu^{5\eta}(p) = \partial_\mu^1(p)$ for $\mu \in \{2,3\}$, and for $\mu \in \{0,1\}$ it is $D_\mu^{\eta,1}\partial_\mu^\eta = \partial_\mu^1$.

(iv) For $n \in \mathbb{Z}^4$ and $x \in T_\eta$ the product $2\pi\eta^{-1}\rho n \cdot x$ is a multiple of 2π, because ρ is an isometry and $\rho x \in \eta\mathbb{Z}^4$; this gives the periodicity of u_k, and that of V follows in the same way from the positions of the bonds. The inclusion holds because $\eta^{-1}\omega\mathbb{Z}[i] \subset \mathbb{Z}[i]$ by Lemma 2.1(iv). The value $u_k(0) = 1$ is the count of Lemma 6.2(i). For $l \in 2\pi\mathbb{Z}^4$ the function $x \mapsto e^{il\cdot x}$ is a character of the group T_η which is trivial on $\mathbb{Z}^4$, and $B_1(0)$ contains one point of every class of T_η modulo $\mathbb{Z}^4$; the sum of a nontrivial character over a finite group vanishes, and the character is trivial only if $l \cdot x \in 2\pi\mathbb{Z}$ for every $x \in T_\eta$, that is, $l \in 2\pi\eta^{-1}\rho\mathbb{Z}^4$. Finally $s_{22}(0) = 3$, $s_{23}(0) = 4$, $s_{32}(0) = 4$, $s_{33}(0) = -3$, which are the numbers $5e_\mu \cdot e_\nu'$ of (2.2). *q.e.d.*

Statement (iii) is the identity $Q_k\partial = \partial_1 Q_k'$ in the momentum representation, and (iv) contains the two properties of u_k which Bałaban uses after [2, (1.33), p. 23]. In his setting a momentum p of the fine lattice is written as $p = p' + l$ with p' in the dual of the unit lattice and $l \in 2\pi\mathbb{Z}^4$, the classes of l being taken modulo $2\pi\eta^{-1}\mathbb{Z}^4$. Here they are taken modulo $2\pi\eta^{-1}\rho\mathbb{Z}^4$, and the number of the classes is again η^{-4}.

The bounds on u_k are those of [2, (1.36), p. 24] with one change. For $p \in \mathbb{R}^4$ let q_μ, for $\mu \in \{0,1\}$, be the number with $q_\mu \equiv p_\mu$ modulo $2\pi/\eta$ and $-\pi/\eta < q_\mu \le \pi/\eta$, and for $\mu \in \{2,3\}$ the number with $q_\mu \equiv p_\mu$ modulo $2\pi/(5\eta)$ and $-\pi/(5\eta) < q_\mu \le \pi/(5\eta)$, so that $(p_2, p_3) = (q_2, q_3) + \frac{2\pi}{5\eta} m$ with $m \in \mathbb{Z}[i]$. We call p *central* if $m \in (3+4i)\mathbb{Z}[i]$.

Lemma 7.2 (Bounds on the Symbols). *(i)* $\kappa(0) = 1$; $\kappa(\theta + \frac{2\pi}{5}m) = \kappa(\theta)$ *for* $m \in (3+4i)\mathbb{Z}[i]$; $\kappa(\frac{2\pi}{5}m) = 0$ *for* $m \in \mathbb{Z}[i] \setminus (3+4i)\mathbb{Z}[i]$; *and* $|\kappa(\theta) - \kappa(\theta')| \le 5\sqrt{2}|\theta - \theta'|$.

(ii) For every $p \in \mathbb{R}^4$, *with* q *as above,*

$$|u_k(p)| \le \prod_{\mu=0}^{3} \min\Big(1, \frac{\pi|\partial_\mu^1(p)|}{2|q_\mu|}\Big) \cdot \begin{cases} 1 & \text{if } p \text{ is central,} \\ \min(1, 5\sqrt{2}\,\eta|(q_2,q_3)|) & \text{otherwise,} \end{cases} \tag{7.7}$$

and $|V_{\mu\nu}(p)| \le \frac{4}{5}\min(1, \pi|\partial_\mu^1(p)|/(2|q_\mu|))$ *for* $\mu, \nu \in \{2,3\}$; *a quotient with* $q_\mu = 0$ *is read as* ∞.

(iii) For $p \in [-\pi, \pi]^4$,

$$|u_k(p)| \ge \Big(\frac{2}{\pi}\Big)^4 (1 - 10\pi\eta)\,, \tag{7.8}$$

and the matrix $V^{\mathrm{pl}}(p) \coloneqq (V_{\mu\nu}(p))_{\mu,\nu\in\{2,3\}}$ *satisfies* $|V^{\mathrm{pl}}(p)w| \geq \frac{2}{\pi}(1-32\eta)|w|$ *for every* $w \in \mathbb{C}^2$.

Proof. (i) $\kappa(0) = 1$ because C has 25 elements. For $m \in \mathbb{Z}[i]$ the function $c \mapsto e^{i\frac{2\pi}{5}m\cdot c}$ is a character of the group $\omega\mathbb{Z}[i]$, trivial on $5\mathbb{Z}[i]$, since $m \cdot c \in 5\mathbb{Z}$ for $c \in 5\mathbb{Z}[i]$; and C contains one element of every class of $\omega\mathbb{Z}[i]$ modulo $5\mathbb{Z}[i]$. The character is trivial on $\omega\mathbb{Z}[i]$ if and only if $m \cdot \omega w \in 5\mathbb{Z}$ for every $w \in \mathbb{Z}[i]$, that is, since $m \cdot \omega w = \mathrm{Re}(\bar\omega m\bar w)$, if and only if $\bar\omega m \in 5\mathbb{Z}[i]$, which means $m \in 5\omega\mathbb{Z}[i] = (3+4i)\mathbb{Z}[i]$. If it is trivial, multiplying the terms of κ by it changes nothing, which is the periodicity; if it is not, its sum over the classes vanishes. The last assertion holds because $|c| < 5\sqrt{2}$ for $c \in \mathsf{C}$.

(ii) For a number a with $1/a$ an integer and a real t with $|at| \leq \pi$ we have $|e^{iat}-1|/a = 2|\sin(at/2)|/a \geq 2|t|/\pi$, so that $|D^{a,1}_\mu(p)| = |\partial^1_\mu(p)|/|\partial^a_\mu(q)| \leq \pi|\partial^1_\mu(p)|/(2|q_\mu|)$ for $a = \eta$, $\mu \in \{0,1\}$, and for $a = 5\eta$, $\mu \in \{2,3\}$, because ∂^a_μ has the period $2\pi/a$ in p_μ; and $|D^{a,1}_\mu| \leq 1$ by (7.2). Further $|\kappa| \leq 1$, and if p is not central, then $\kappa(\eta(p_2,p_3)) = \kappa(\eta(q_2,q_3)+\frac{2\pi}{5}m)-\kappa(\frac{2\pi}{5}m)$ by (i), which is at most $5\sqrt{2}\,\eta|(q_2,q_3)|$ in modulus. With (7.4) this is (7.7), and the bound on V follows from (7.5) and $|s_{\mu\nu}| \leq 4$.

(iii) For $|t| \leq \pi$ and $a \leq 1$ we have $|e^{it}-1| = 2|\sin(t/2)| \geq 2|t|/\pi$ and $|e^{iat}-1|/a \leq |t|$, so that $|D^{a,1}_\mu(p)| \geq 2/\pi$; and $|\kappa(\eta(p_2,p_3))-1| \leq 5\sqrt{2}\,\eta|(p_2,p_3)| \leq 10\pi\eta$ by (i). This gives (7.8). Let R_0 be the matrix with the entries $\frac{1}{5}s_{\mu\nu}(0) = e_\mu \cdot e'_\nu$, which is orthogonal by (2.2). Each $s_{\mu\nu}$ has at most four terms $\pm e^{i\theta\cdot c}$ with $|c| \leq 3\sqrt{2}$, so that $|\frac{1}{5}s_{\mu\nu}(\theta) - \frac{1}{5}s_{\mu\nu}(0)| \leq \frac{12\sqrt{2}}{5}|\theta|$, and the norm of the matrix of these differences is at most twice this number. For $\theta = \eta(p_2,p_3)$, $|\theta| \leq \sqrt{2}\,\pi\eta$, this is at most $\frac{48\pi}{5}\eta \leq 32\eta$. Since $V^{\mathrm{pl}}(p)$ is the product of the diagonal matrix of the numbers $D^{5\eta,1}_\mu(p)$, of modulus at least $2/\pi$, with the matrix $\frac{1}{5}s(\theta)$, the assertion follows. *q.e.d.*

The first product in (7.7) is Bałaban's bound $\prod_\mu |p'_\mu|/|p'_\mu + l_\mu|$, with the difference that in the turned directions q_μ is the representative of p_μ in an interval of the length $2\pi/(5\eta)$ rather than $2\pi/\eta$. Each value of q therefore belongs to 25 of the classes of l; one of them is central, and on the other 24 the symbol carries the additional factor $\eta|q|$, while $\Delta(p)$ is of the order of η^{-2} there.

As a first application we give the bound on the operator $\phi = Q_k\Delta^{-1}Q_k^*$ of [2, (1.58), (1.62), pp. 27–28], which in the tilted run is a matrix in the turned directions. For p' in the dual of the unit lattice let

$$\phi_{\mu\mu'}(p') \coloneqq \sum_l \frac{|u_k(p'+l)|^2}{\Delta(p'+l)} \sum_\nu V_{\mu\nu}(p'+l)\,\overline{V_{\mu'\nu}(p'+l)}\,, \tag{7.9}$$

the sum over the classes of $l \in 2\pi\mathbb{Z}^4$ modulo $2\pi\eta^{-1}\rho\mathbb{Z}^4$. By Lemma 7.1(i) and (ii) this is the momentum representation of ϕ: $(\widetilde{\phi\omega})_\mu(p') = \sum_{\mu'} \phi_{\mu\mu'}(p')\tilde\omega_{\mu'}(p')$. It is diagonal for $\mu \in \{0,1\}$, where it is the function ϕ_μ of [2, (1.62)] with the tilted u_k, and a Hermitian matrix on the indices $\{2,3\}$.

Lemma 7.3 (The Matrix ϕ)**.** *Let* $32\eta \leq \frac{1}{2}$. *There are constants* $0 < \gamma_0 \leq \gamma_1$, *which do not depend on* k *and* L, *such that* $\gamma_0|w|^2 \leq \Delta_0(p')\sum_{\mu,\mu'} \bar w_\mu \phi_{\mu\mu'}(p') w_{\mu'} \leq \gamma_1|w|^2$ *for every* $p' \in [-\pi,\pi]^4$, $p' \neq 0$, *and every* $w \in \mathbb{C}^4$.

Proof. The quadratic form is $\sum_l |u_k(p'+l)|^2\Delta_0(p')\Delta(p'+l)^{-1}|V(p'+l)^*w|^2$, a sum of non-negative terms. For the lower bound we keep the term $l = 0$. Since $|e^{it}-1| \leq |t|$, we have

$\Delta(p') \le \sum_\nu |p' \cdot e'_\nu|^2 = |p'|^2$, and $\Delta_0(p') \ge (4/\pi^2)|p'|^2$ because $|e^{it} - 1| \ge 2|t|/\pi$ for $|t| \le \pi$. With Lemma 7.2(iii), applied to V^{pl} and to the adjoint of this invertible matrix, and with $|D^{\eta,1}_\mu(p')| \ge 2/\pi$ for $\mu \in \{0,1\}$, the term is at least $(4/\pi^2)(2/\pi)^8(1-10\pi\eta)^2(2/\pi)^2(1-32\eta)^2|w|^2$, and $10\pi\eta \le 32\eta \le \frac{1}{2}$.

For the upper bound, $|V(p)^* w| \le \frac{8}{5}|w|$ by Lemma 7.2(ii). Let $\hat{p}$ be the representative of $p' + l$ with $|\eta\hat{p} \cdot e'_\nu| \le \pi$ for every ν. Then $\Delta(p' + l) = \Delta(\hat{p}) \ge (4/\pi^2)|\hat{p}|^2$, and $\Delta_0(p') \le |p'|^2$. For $l = 0$ we have $\hat{p} = p'$ and the term is at most $(\pi^2/4)(8/5)^2|w|^2$. For $l \ne 0$ the difference $\hat{p} - p'$ is a nonzero element of $2\pi\mathbb{Z}^4$ by Lemma 7.1(iv), so that one coordinate of $\hat{p}$ is at least π in modulus, and $\Delta_0(p')/\Delta(p' + l) \le (\pi^2/4)|p'|^2/\pi^2 \le \pi^2$. By (7.7), $|u_k(p' + l)| \le \prod_\mu \min(1, \pi/|q_\mu|)$, since $|\partial^1_\mu| \le 2$, and each value of q, which is of the form $q_\mu = p'_\mu + 2\pi n_\mu$ with integers n_μ, belongs to at most 25 classes of l. Hence the terms with $l \ne 0$ add up to at most $25\pi^2(8/5)^2 \prod_\mu \sum_{n\in\mathbb{Z}} \min(1, \pi/|p'_\mu + 2\pi n|)^2|w|^2 \le 25\pi^2(8/5)^2(1 + \pi^2/4)^4|w|^2$, because $|p'_\mu + 2\pi n| \ge \pi(2|n| - 1)$ for $n \ne 0$ and $\sum_{n\ge1}(2n-1)^{-2} = \pi^2/8$. *q.e.d.*

8 The Cells

Several of Bałaban's arguments are local. They use paths of fine bonds inside a block, and they use the lower bound for the Laplace operator of a block with Neumann conditions on the functions of mean zero, which for a cube of the fine lattice is explicit [1, (2.27), p. 580]. In the tilted run a block is a cube of the coarse lattice filled with points of the turned lattice, and neither the paths nor the bound are evident. Both follow from the description of a block as a union of *cells*, the blocks $B_{5\eta}(z)$, $z \in 5\eta\mathbb{Z}^4$, of the scale 5η: by Lemma 6.2(i) the cells of a block form a cubic array with the axes e_μ, and each of them is a translate of the same set of 5^4 points, which we now describe.

Lemma 8.1 (The Cell)**.** *Let* $\mathsf{R} \subset \mathbb{Z}^2$ *be the set of the pairs* (m, n) *with* $m \in \{3, 4\}$ *for* $n = -3$, $2 \le m \le 5$ *for* $n = -2$, $1 \le m \le 6$ *for* $n = -1$, $0 \le m \le 6$ *for* $n = 0$, $2 \le m \le 5$ *for* $n = 1$, *and* $m \in \{3, 4\}$ *for* $n = 2$.

(i) $\mathsf{C} = \{me'_2 - ne'_3 \,|\, (m, n) \in \mathsf{R}\}$, *the vectors* e'_2 *and* e'_3 *being read in the plane. Hence the cell of* $z \in 5\eta\mathbb{Z}^4$ *consists of the points* $z + \eta(n_0 e_0 + n_1 e_1 + me'_2 - ne'_3)$ *with* $n_0, n_1 \in \{0, \dots, 4\}$ *and* $(m, n) \in \mathsf{R}$.

(ii) Every row $\{m \,|\, (m, n) \in \mathsf{R}\}$ *is an interval which contains* 3 *and* 4.

(iii) The points $z + \eta(3e'_2 + 4e'_3)$ *and* $z + \eta(4e'_2 - 3e'_3)$ *are the corners* $z + 5\eta e_2$ *and* $z + 5\eta e_3$ *of the neighbouring cells.*

Proof. (i) An element of $\omega\mathbb{Z}[i]$ is $c = \omega(m + ni) = m\omega + n\omega i$ with integers m, n, and ω and $\omega i = (-4 + 3i)/5$ are the vectors e'_2 and $-e'_3$ of the plane by (2.2). Its real and imaginary parts are $(3m - 4n)/5$ and $(4m + 3n)/5$, so that $c \in \mathsf{C}$ if and only if $0 \le 3m - 4n < 25$ and $0 \le 4m + 3n < 25$, that is, $\max(\frac{4}{3}n, -\frac{3}{4}n) \le m < \min(\frac{25+4n}{3}, \frac{25-3n}{4})$. For $n = -3, \dots, 2$ these bounds are $\frac{9}{4} \le m < \frac{13}{3}$, $\frac{3}{2} \le m < \frac{17}{3}$, $\frac{3}{4} \le m < 7$, $0 \le m < \frac{25}{4}$, $\frac{4}{3} \le m < \frac{11}{2}$, and $\frac{8}{3} \le m < \frac{19}{4}$, which give the rows of R. For $n \ge 3$ they require $m \ge 4$ and $m < 4$, and for $n \le -4$ they require $m \ge 3$ and $m < 3$. (ii) is read off from the list, and (iii) is (2.2). *q.e.d.*

By (ii) and (iii) two neighbouring corners of cells are joined by short paths which stay in the cell of the first: for the direction e_2 three steps $\eta e'_2$ followed by four steps $\eta e'_3$, through the points $(m, n) = (1, 0), (2, 0), (3, 0), (3, -1), (3, -2), (3, -3)$ of R, and for the direction e_3 four steps $\eta e'_2$ followed by three steps $-\eta e'_3$, through $(1, 0), \dots, (4, 0), (4, 1), (4, 2)$. We call these

paths, and the straight paths of five bonds in the directions e_0 and e_1, the *connectors* from z.

Definition 8.2 (Contours in a Block)**.** Let a be a scale, $y \in a\mathbb{Z}^4$, and $x \in B_a(y)$, let $z \in 5\eta\mathbb{Z}^4$ be the corner of the cell of x, and write $x = z+\eta(n_0e_0+n_1e_1+me_2'-ne_3')$ as in Lemma 8.1(i). The contour $\Gamma^a_{y,x}$ is the following path of bonds of T_η from y to x. It first goes from y to z through corners of cells, changing the coordinates in the order $\mu = 3, 2, 1, 0$, each step from a corner to the next along the connector. It then makes n_0 steps ηe_0 and n_1 steps ηe_1, and finally it goes through the points of the cell with the plane coordinates $(1, 0)$, $(2, 0)$, $(3, 0)$, from there along the column $m = 3$ to $(3, n)$, and along the row n to (m, n).

By Lemma 8.1 every site of $\Gamma^a_{y,x}$ lies in $B_a(y)$, the part of the contour after z lies in the cell of x and has at most $4 + 4 + 9$ bonds, and the contour has at most $\frac{7}{5} \cdot 4a/\eta + 17$ bonds. These contours take the place of [2, (1.7), p. 18] and of the contours $\Gamma^{(k)}_{y,x}$ of [1, (1.4), p. 572], which are required to lie in the block.

Lemma 8.3 (A Poincaré Inequality)**.** *For every scale a, every $y \in a\mathbb{Z}^4$, and every function λ on $B_a(y)$ with $\sum_{x\in B_a(y)} \lambda(x) = 0$,*

$$\sum_{x\in B_a(y)} |\lambda(x)|^2 \le C_P\Big(\frac{a}{\eta}\Big)^2 \sum_b |\lambda(b_+) - \lambda(b_-)|^2 \,, \qquad C_P := 1550 \,, \tag{8.1}$$

the sum on the right over the bonds $b = \langle b_-, b_+\rangle$ of T_η with both ends in $B_a(y)$.

Proof. Write $\delta\lambda(b) := \lambda(b_+) - \lambda(b_-)$, and let $N := a/(5\eta)$, so that $B_a(y)$ is the union of N^4 cells, whose corners form a cube of N^4 points of $5\eta\mathbb{Z}^4$. For a point x of a cell with the corner z, the last part of the contour of Definition 8.2 is a path of at most 17 bonds inside the cell, so that $|\lambda(x)-\lambda(z)|^2$ is at most 17 times the sum of $|\delta\lambda|^2$ over the bonds of the cell; summing over the 625 points,

$$\sum_x |\lambda(x) - \lambda(z)|^2 \le 625 \cdot 17 \sum_b |\delta\lambda(b)|^2 \,, \tag{8.2}$$

the sums over the points and the bonds of the cell. Let Λ be the restriction of λ to the corners, and $\bar\Lambda$ its mean. For two neighbouring corners z and $z+5\eta e_\mu$ in $B_a(y)$ the connector has at most 7 bonds, all with both ends in $B_a(y)$, so that $|\Lambda(z + 5\eta e_\mu) - \Lambda(z)|^2$ is at most 7 times the sum of $|\delta\lambda|^2$ over its bonds, and a bond of T_η lies on at most four connectors, those from the corner of its cell. For a function f on $\{0, \dots, N-1\}$ with the mean $\bar f$ we have $\sum_i |f_i - \bar f|^2 = \frac{1}{2N}\sum_{i,j} |f_i - f_j|^2$ and $|f_i - f_j|^2 \le N \sum_m |f_{m+1} - f_m|^2$, hence $\sum_i |f_i - \bar f|^2 \le \frac{1}{2}N^2 \sum_m |f_{m+1} - f_m|^2$. On the cube of the corners, let E_μ be the average over the coordinate μ; then $\Lambda - \bar\Lambda = \sum_{\mu=0}^3 E_0 \cdots E_{\mu-1}(1 - E_\mu)\Lambda$ is a sum of mutually orthogonal functions, the operators E_μ are contractions, and the inequality in one variable applies to $(1 - E_\mu)\Lambda$ on every line. Therefore

$$\sum_z |\Lambda(z) - \bar\Lambda|^2 \le \frac{N^2}{2} \sum_\mu \sum_z |\Lambda(z + 5\eta e_\mu) - \Lambda(z)|^2 \le \frac{N^2}{2} \cdot 7 \cdot 4 \sum_b |\delta\lambda(b)|^2 \,, \tag{8.3}$$

the last sum over the bonds with both ends in $B_a(y)$. Since λ has the mean zero, $\sum_x |\lambda(x)|^2 \le \sum_x |\lambda(x) - \bar\Lambda|^2$, and $|\lambda(x) - \bar\Lambda|^2 \le 2|\lambda(x) - \lambda(z)|^2 + 2|\Lambda(z) - \bar\Lambda|^2$ for x in the cell of z. With (8.2) and (8.3) the left side of (8.1) is at most $(2 \cdot 625 \cdot 17 + 2 \cdot 625 \cdot 14N^2) \sum_b |\delta\lambda(b)|^2 \le 38750\, N^2 \sum_b |\delta\lambda(b)|^2$, and $38750/25 = 1550$. *q.e.d.*

In the notation of [1, (2.27)], the lemma for $a = 1$ says that the Laplace operator of a unit block with Neumann conditions is at least C_P^{-1} on the functions of mean zero, where for a cube of $\eta\mathbb{Z}^4$ the bound is π^2.

9 Bounds in Position Space

Bałaban obtains the bounds of his Proposition 1.1 and the inequality [2, (1.67)] by inspection of the momentum representation, in which every operator is a scalar in the index of the field. With the matrices of Section 7 this inspection becomes laborious, and we prove the same statements in another way: four inequalities for the tilted average, proved by counting paths, the Poincaré inequality of Lemma 8.3, and the momentum representation of two scalar operators only. In this section the fine lattice is the torus T_η/Γ' with a group of periods $\Gamma' \subset \mathbb{Z}^4$, the scalar products are the sums over the points, the positive bonds, or the plaquettes, with the weight η^4 on the fine lattice and 1 on the unit lattice, and $\|\cdot\|$ are the corresponding norms. For a field A and a positive direction ν we write $(\partial'_\nu A)(b) \coloneqq \eta^{-1}(A(b+\eta e'_\nu) - A(b))$, so that $\langle A, \Delta A\rangle = \sum_\nu \|\partial'_\nu A\|^2 = \|\partial A\|^2 + \|\partial^* A\|^2$ by [2, (1.21)], and ∂_1, ∂_1^* are the operations of the unit lattice.

Lemma 9.1 (Four Inequalities for the Average)**.** *For every real field A on T_η/Γ',*

(i) $\|Q_k A\|^2 \le 2\|A\|^2$;

(ii) $\|\partial_1^* Q_k A\|^2 \le 62\,\langle A, \Delta A\rangle$;

(iii) $\|\partial_1 Q_k A\|^2 \le 24\|\partial A\|^2$;

(iv) $\|A\|^2 \le (2C_P + 64)\langle A, \Delta A\rangle + 4\|Q_k A\|^2$.

Proof. Let n_μ be the number of the bonds of a staircase σ^1_μ, so that $n_\mu \le \frac{7}{5\eta}$, and recall from the proof of Lemma 6.4(ii) that a bond of T_η lies on at most $\frac{7}{5\eta}$ of the staircases $\sigma^1_\mu(x)$, $x \in T_\eta$, $\mu = 0, \dots, 3$. We use the Cauchy–Schwarz inequality in the form $|\sum_{t\in\mathcal{T}} c_t|^2 \le |\mathcal{T}| \sum_t |c_t|^2$.

(i) By (6.3), $(Q_k A)(\langle y, y + e_\mu\rangle)$ is η^5 times a sum of the values of A over the $\eta^{-4} n_\mu$ pairs of a point $x \in B_1(y)$ and a bond of $\sigma^1_\mu(x)$. Its square is therefore at most $\eta^{10}\eta^{-4} n_\mu$ times the sum of $|A(b)|^2$ over these pairs, and in the sum over y and μ every bond occurs at most $\frac{7}{5\eta}$ times. Hence $\|Q_k A\|^2 \le \frac{49}{25}\eta^4 \sum_b |A(b)|^2$.

(ii) We have $(\partial_1^* B)(y) = \sum_\mu (B(\langle y - e_\mu, y\rangle) - B(\langle y, y + e_\mu\rangle))$, and $(Q_k A)(\langle y, y + e_\mu\rangle) - (Q_k A)(\langle y - e_\mu, y\rangle) = \eta^5 \sum_{x\in B_1(y)} (A(\sigma^1_\mu(x)) - A(\sigma^1_\mu(x - e_\mu)))$. The staircase $\sigma^1_\mu(x)$ is the translate of $\sigma^1_\mu(x - e_\mu)$ by e_μ. For a bond b the difference $A(b + e_\mu) - A(b)$ is the sum of the n_μ differences of A between the translates of b by two consecutive sites of $\sigma^1_\mu(0)$, each of which is $\pm\eta(\partial'_\nu A)(b'')$ for a translate b'' of b and a direction ν. The difference of the two averages is therefore $\eta^5 \cdot \eta$ times a sum of $\eta^{-4} n_\mu^2$ values $\pm(\partial'_\nu A)(b'')$, and in the sum over y every pair (b'', ν) occurs at most n_μ^2 times. This gives $\sum_y |\cdot|^2 \le \eta^{12}\eta^{-4} n_\mu^4 \sum_{b'',\nu} |(\partial'_\nu A)(b'')|^2 \le (\frac{7}{5})^4 \langle A, \Delta A\rangle$ for every μ, and $|(\partial_1^* Q_k A)(y)|^2$ is at most four times the sum of the squares of the four differences; $16(\frac{7}{5})^4 \le 62$.

(iii) Let P be the plaquette of the unit lattice with the corners y, $y + e_\mu$, $y + e_\mu + e_\nu$, $y + e_\nu$. Since $B_1(y + e_\mu) = B_1(y) + e_\mu$, we have $(\partial_1 Q_k A)(P) = \eta^5 \sum_{x\in B_1(y)} A(\ell(x))$ with the closed path $\ell(x)$ which runs along $\sigma^1_\mu(x)$ and $\sigma^1_\nu(x + e_\mu)$ and back along $\sigma^1_\mu(x + e_\nu)$ and $\sigma^1_\nu(x)$. Let $s_0 = 0, s_1, \dots, s_{n_\mu} = e_\mu$ be the sites of $\sigma^1_\mu(0)$ and $s'_0, \dots, s'_{n_\nu}$ those of $\sigma^1_\nu(0)$. The points $x + s_i + s'_j$ form a mesh whose boundary is $\ell(x)$, and the sum of A over $\ell(x)$ is the sum over i and j of the sums of A over the closed paths through $x + s_i + s'_j$, $x + s_{i+1} + s'_j$, $x + s_{i+1} + s'_{j+1}$, $x + s_i + s'_{j+1}$. If the two steps $s_{i+1} - s_i$ and $s'_{j+1} - s'_j$ are parallel, such a path runs through each of its bonds once in each sense and contributes nothing; otherwise it is the boundary of a plaquette p of T_η and contributes $\pm\eta(\partial A)(p)$. Hence $(\partial_1 Q_k A)(P)$ is $\eta^5 \cdot \eta$ times a sum

of at most $\eta^{-4} n_\mu n_\nu$ values $\pm(\partial A)(p)$, and in the sum over y every plaquette occurs at most $n_\mu n_\nu$ times. This gives $\sum_y |(\partial_1 Q_k A)(P)|^2 \le (\frac{7}{5})^4 \|\partial A\|^2$ for each of the six pairs $\mu < \nu$, and $6(\frac{7}{5})^4 \le 24$.

(iv) Let $A_\nu(x) \coloneqq A(\langle x, x+\eta e'_\nu\rangle)$, and let $\bar{A}_\nu(y)$ be the mean of A_ν over $B_1(y)$. Lemma 8.3 with $a = 1$, applied to $A_\nu - \bar{A}_\nu(y)$ on every block, gives $\sum_\nu \sum_x \eta^4 |A_\nu(x) - \bar{A}_\nu(y(x))|^2 \le C_P \langle A, \Delta A\rangle$, where $y(x)$ is the corner of the block of x, because the differences of A_ν along the bonds of a block are among the numbers $\eta(\partial'_{\nu'} A)(b)$. Since a block has η^{-4} points, $\|A\|^2 \le 2C_P \langle A, \Delta A\rangle + 2\sum_y \sum_\nu |\bar{A}_\nu(y)|^2$. By Lemma 6.4(ii) the average of the constant field with the values $\bar{A}_\nu(y)$ is $\sum_\nu (e_\mu \cdot e'_\nu)\bar{A}_\nu(y)$, so that $(Q_k A)(\langle y, y+e_\mu\rangle) - \sum_\nu (e_\mu \cdot e'_\nu)\bar{A}_\nu(y)$ is η^5 times the sum, over $x \in B_1(y)$ and over the bonds of $\sigma^1_\mu(x)$, of the differences between the value of A on the bond and the value of the same component at x. Each of these differences is a sum of at most n_μ terms $\pm\eta(\partial'_{\nu'} A)(b'')$ along the staircase, and the counting of (ii) bounds the sum over y of the squares by $(\frac{7}{5})^4 \langle A, \Delta A\rangle$ for every μ. The matrix $(e_\mu \cdot e'_\nu)$ is orthogonal, so that $\sum_\nu |\bar{A}_\nu(y)|^2 \le 2\sum_\mu |(Q_k A)(\langle y, y+e_\mu\rangle)|^2 + 2\sum_\mu |\cdot|^2$ with the differences just bounded, and $\sum_y \sum_\nu |\bar{A}_\nu(y)|^2 \le 2\|Q_k A\|^2 + 8(\frac{7}{5})^4 \langle A, \Delta A\rangle$. Since $16(\frac{7}{5})^4 \le 64$, this is (iv). *q.e.d.*

The two scalar operators are those of [2, (1.38), (1.58)] for functions. As there, Δ^{-1} denotes the inverse of the Laplace operator $\Delta = \partial^* \partial$ of T_η/Γ' on the functions orthogonal to the constants, and $\Delta_0 \coloneqq \partial_1^* \partial_1$ is the Laplace operator of the unit lattice.

Lemma 9.2 (Two Scalar Operators)**.** *Let $10\pi\eta \le \frac{1}{2}$. On the functions of the unit lattice which are orthogonal to the constants, the operators $\mathsf{S} \coloneqq Q'_k \Delta^{-2} Q'^*_k$ and $\mathsf{T} \coloneqq Q'_k \Delta^{-1} Q'^*_k$ satisfy*

$$\mathsf{S} \ge c_{\mathsf{S}} \Delta_0^{-2} , \qquad \mathsf{T} \le C_{\mathsf{T}} \Delta_0^{-1} , \qquad c_{\mathsf{S}} \coloneqq \tfrac{1}{4}(\tfrac{2}{\pi})^{12} , \quad C_{\mathsf{T}} \coloneqq \tfrac{\pi^2}{4} + 100(1 + \tfrac{\pi^2}{4})^4 . \tag{9.1}$$

Proof. By Lemma 7.1(i) and its adjoint, both operators are multiplications in the momentum representation of the unit lattice, by $\mathsf{S}(p') = \sum_l |u_k(p'+l)|^2 \Delta(p'+l)^{-2}$ and $\mathsf{T}(p') = \sum_l |u_k(p'+l)|^2 \Delta(p'+l)^{-1}$, for $p' \in [-\pi, \pi]^4$, $p' \ne 0$. As in the proof of Lemma 7.3, $(4/\pi^2)|p'|^2 \le \Delta_0(p') \le |p'|^2$ and $(4/\pi^2)|p'|^2 \le \Delta(p') \le |p'|^2$. The term $l = 0$ of S is at least $(2/\pi)^8 (1 - 10\pi\eta)^2 |p'|^{-4}$ by (7.8), and $|p'|^{-4} \ge (4/\pi^2)^2 \Delta_0(p')^{-2}$. The term $l = 0$ of T is at most $(\pi^2/4)|p'|^{-2} \le (\pi^2/4)\Delta_0(p')^{-1}$. For $l \ne 0$ we have $\Delta(p'+l) \ge 4$ and $\sum_{l \ne 0} |u_k(p'+l)|^2 \le 25(1 + \pi^2/4)^4$ by the proof of Lemma 7.3, and $1 \le 16\Delta_0(p')^{-1}$. *q.e.d.*

Proposition 9.3 (Proposition 1.1 and the Inequality (1.67) for the Tilted Run)**.** *Let $32\eta \le \frac{1}{2}$ and $a > 0$. There are constants $\gamma_0 > 0$ and C, which depend on a and do not depend on k, L, and the torus, such that the operator $\Delta_a = \Delta - \partial P \partial^* + a Q_k^* Q_k$ of [2, (1.69)] and its inverse G satisfy*

$$\Delta_a \ge \gamma_0 (\Delta + I) , \tag{9.2}$$

$$\|GJ\|, \ \|\nabla GJ\|, \ \|G\nabla^* J\|, \ \|\nabla G \nabla^* J\| \le \gamma_0^{-1} \|J\| , \qquad \|\nabla\nabla GJ\|, \ \|G\nabla^*\nabla^* J\| \le C\|J\| , \tag{9.3}$$

where $\nabla A \coloneqq (\partial'_\nu A)_\nu$, and such that the action Δ_k of [2, (1.65)] satisfies

$$\tfrac{1}{24} \|\partial_1 B\|^2 \le \langle B, \Delta_k B\rangle \le \gamma_\phi^{-1} \|\partial_1 B\|^2 , \tag{9.4}$$

where γ_ϕ denotes the lower constant γ_0 of Lemma 7.3.

Proof. We use the operator identities of [2], which hold in the tilted run by Section 10: $R = I - P$ is the orthogonal projection onto $\Delta N(Q'_k)$, where $N(Q'_k)$ is the null space of

Q'_k; $P = \Delta^{-1} Q'^*_k \mathsf{S}^{-1} Q'_k \Delta^{-1}$ on the functions orthogonal to the constants; and $\langle A, \Delta_a A\rangle = \|\partial A\|^2 + \|R\partial^* A\|^2 + a\|Q_k A\|^2$.

A gauge transformation. Let A be a field. There is $\lambda \in N(Q'_k)$ with $\Delta\lambda = R\partial^* A$, and $A_s := A - \partial\lambda$ satisfies $\partial A_s = \partial A$, $Q_k A_s = Q_k A - \partial_1 Q'_k \lambda = Q_k A$ by Lemma 6.4(i), and $\partial^* A_s = \partial^* A - \Delta\lambda = P\partial^* A$. Hence

$$\langle A, \Delta_a A\rangle = \|\partial A_s\|^2 + a\|Q_k A_s\|^2 + \|\Delta\lambda\|^2 \,. \tag{9.5}$$

The gauge part. Since $\partial\partial\lambda = 0$, we have $\langle \partial\lambda, \Delta\partial\lambda\rangle = \|\partial^*\partial\lambda\|^2 = \|\Delta\lambda\|^2$. The function λ has the mean zero on every unit block, so that $\|\lambda\|^2 \le C_P\|\partial\lambda\|^2$ by Lemma 8.3 with $a = 1$, and $\|\partial\lambda\|^2 = \langle\lambda, \Delta\lambda\rangle \le \|\lambda\|\|\Delta\lambda\|$; together, $\|\partial\lambda\|^2 \le C_P\|\Delta\lambda\|^2$.

The divergence on the slice. Put $f := \partial^* A_s$, so that $f = Pf$ is orthogonal to the constants, and $\mu := Q'_k \Delta^{-1} f$. Then $\|f\|^2 = \langle f, Pf\rangle = \langle \mu, \mathsf{S}^{-1}\mu\rangle \le c_{\mathsf{S}}^{-1}\|\Delta_0\mu\|^2$ by Lemma 9.2. The field $A^T := A_s - \partial\Delta^{-1} f$ satisfies $\partial^* A^T = 0$ and $\partial A^T = \partial A_s$, and $\partial_1\mu = Q_k\partial\Delta^{-1} f = Q_k A_s - Q_k A^T$ by Lemma 6.4(i). Hence $\Delta_0\mu = \partial_1^* Q_k A_s - \partial_1^* Q_k A^T$, where $\|\partial_1^* Q_k A_s\|^2 \le 16\|Q_k A_s\|^2$, each bond of the unit lattice entering ∂_1^* at its two ends, and $\|\partial_1^* Q_k A^T\|^2 \le 62\langle A^T, \Delta A^T\rangle = 62\|\partial A_s\|^2$ by Lemma 9.1(ii). Therefore

$$\|\partial^* A_s\|^2 \le C_4(\|\partial A_s\|^2 + \|Q_k A_s\|^2) \,, \qquad C_4 := 124\, c_{\mathsf{S}}^{-1} \,. \tag{9.6}$$

The lower bound. By (9.6), $\langle A_s, \Delta A_s\rangle \le (1 + C_4)(\|\partial A_s\|^2 + \|Q_k A_s\|^2)$, and by Lemma 9.1(iv), $\|A_s\|^2 \le (2C_P + 64)\langle A_s, \Delta A_s\rangle + 4\|Q_k A_s\|^2$. With the bounds on the gauge part and $\langle A, (\Delta + I)A\rangle \le 2\langle A_s, (\Delta + I)A_s\rangle + 2\langle\partial\lambda, (\Delta + I)\partial\lambda\rangle$, the identity (9.5) gives (9.2) with

$$\gamma_0^{-1} := 2\max\Big(\frac{(2C_P + 65)(1 + C_4) + 4}{\min(1, a)},\ 1 + C_P\Big) \,. \tag{9.7}$$

The bounds on G. By (9.2), Δ_a is invertible and $(\Delta + I)^{1/2} G (\Delta + I)^{1/2}$ has the norm at most γ_0^{-1}; since $\|\nabla A\|^2 = \langle A, \Delta A\rangle$, this is the first half of (9.3). On the torus the operators ∂'_ν commute with each other and with their adjoints, so that $\|\nabla\nabla A\|^2 = \|\Delta A\|^2$, and $\Delta G = I + \partial P\partial^* G - aQ_k^* Q_k G$. Here $\|Q_k^* Q_k G J\| \le 2\gamma_0^{-1}\|J\|$ by Lemma 9.1(i). For a function g orthogonal to the constants, $Pg = \Delta^{-1} Q'^*_k \omega$ with $\omega = \mathsf{S}^{-1} Q'_k \Delta^{-1} g$, so that $\|\partial Pg\|^2 = \langle Pg, \Delta Pg\rangle = \langle\omega, \mathsf{T}\omega\rangle \le C_{\mathsf{T}}\langle\omega, \Delta_0^{-1}\omega\rangle$, while $\|g\|^2 \ge \|Pg\|^2 = \langle\omega, \mathsf{S}\omega\rangle \ge c_{\mathsf{S}}\langle\omega, \Delta_0^{-2}\omega\rangle \ge \frac{1}{16} c_{\mathsf{S}}\langle\omega, \Delta_0^{-1}\omega\rangle$, because $\Delta_0 \le 16$. With $g = \partial^* GJ$, $\|g\| \le \|\nabla GJ\|$, this gives $\|\Delta GJ\| \le (1 + 2a\gamma_0^{-1} + (16C_{\mathsf{T}}/c_{\mathsf{S}})^{1/2}\gamma_0^{-1})\|J\|$, and the last bound of (9.3) follows by taking adjoints.

The action. By [2, (1.47)–(1.65)], $\langle B, \Delta_k B\rangle = \|\partial H_k B\|^2$ is the minimum of $\|\partial A\|^2$ over the fields with $Q_k A = B$ and $R\partial^* A = 0$, and by the gauge transformation above the second condition may be dropped. Lemma 9.1(iii) for $A = H_k B$ is the lower bound of (9.4). For the upper bound write $B = B_0 + B'$ with B_0 constant and B' orthogonal to the constants, and $B' = B^T + \partial_1\nu$ with $\nu := \Delta_0^{-1}\partial_1^* B'$, so that $\partial_1^* B^T = 0$ and $\langle B^T, \Delta_0 B^T\rangle = \|\partial_1 B^T\|^2 = \|\partial_1 B\|^2$. Let $\phi = Q_k\Delta^{-1} Q_k^*$, whose momentum representation is (7.9); by Lemma 7.3, $\phi^{-1} \le \gamma_\phi^{-1}\Delta_0$ on the fields orthogonal to the constants. The field $A := Q_k^* B_0 + \Delta^{-1} Q_k^*\phi^{-1} B^T + \partial Q'^*_k\nu$ satisfies $Q_k A = B_0 + B^T + \partial_1 Q'_k Q'^*_k\nu = B$, by Lemma 6.4 and because $Q'_k Q'^*_k = I$, and $\partial A = \partial\Delta^{-1} Q_k^*\phi^{-1} B^T$. Hence $\langle B, \Delta_k B\rangle \le \|\partial A\|^2 \le \langle\Delta^{-1} Q_k^*\phi^{-1} B^T, Q_k^*\phi^{-1} B^T\rangle = \langle B^T, \phi^{-1} B^T\rangle \le \gamma_\phi^{-1}\|\partial_1 B\|^2$. *q.e.d.*

The proposition has the content of [2, Proposition 1.1, (1.89), (1.90), and (1.67)]; the two bounds on second derivatives hold with a constant other than γ_0^{-1}.

10 The Modifications in the First Paper on the Propagators

We go through [2] in its order. The paper treats the Gaussian theory of one real field, with the action $\frac{1}{2}\langle \partial A, \partial A\rangle$ on the fine lattice, and it derives the propagators of the later papers from the averages. In the tilted run its fine lattice is T_η of (6.1), on a torus T_η/Γ' with a group of periods $\Gamma' \subset \mathbb{Z}^4$ in place of the box of [2, p. 17]; the operators ∂, ∂^*, and Δ are those of this lattice, whose bonds have the directions e'_ν, and a field has the components A_ν in these directions, while the fields B and the multipliers ω on the unit lattice have the components in the directions e_μ.

Section A, (1.1)–(1.15). The action [2, (1.3), (1.5)] and its gauge invariance [2, (1.4)] concern the fine lattice alone and are unchanged. The blocks [2, (1.6)] are those of Definition 6.1(i), and the average [2, (1.8), (1.11)] is replaced by Definition 6.1(iv); the transformation laws [2, (1.9), (1.13)] are Lemma 6.4(i). The contours [2, (1.7)] enter the axial gauge condition [2, (1.10)], which requires for every point x of a block a path of fine bonds from the corner y to x; we take the path which changes the coordinates of $x - y$ in the basis $\eta e'_\nu$ in the order $\nu = 3, 2, 1, 0$. Such a path may leave the block, by less than its side, which is of no consequence in [2, (1.10)–(1.15)]: these use only that the conditions $A(\Gamma_{y,x}) = 0$ fix the gauge transformations which vanish at the corners, one condition for each point $x \neq y$.

Section B, (1.16)–(1.20). The computation [2, (1.16)] of a composition is Lemma 6.3: the average of the first step of the tilted run is $Q_{L^{-1}}$ in the unit of the lattice reached after k steps, the later ones are Bałaban's, and the composition is Q_k. The formula [2, (1.20)] is Lemma 6.4(i).

Sections C to E, the operator identities. The derivations of [2, (1.21)–(1.28), (1.38)–(1.44), (1.46)–(1.60), (1.64), (1.65), (1.68)–(1.82), (1.91)–(1.107)] are calculations with the operators ∂, ∂^*, Δ of the fine lattice, with Q_k, Q'_k, their adjoints, and with ∂_1 on the unit lattice. They use the following properties and no others: the decomposition [2, (1.21)] of the action, which holds on every lattice with four orthogonal directions of bonds and hence on T_η; the identity [2, (1.55)], which is Lemma 6.4(i); that Q'_k maps constants to constants and their orthogonal complement to the orthogonal complement; and, in [2, (1.57)–(1.59), (1.74), (1.82)], that Q_k and Q_k^* map the constant fields to the constant fields with $Q_k^* Q_k = 1$ on them. The last two properties are Lemma 6.4(ii). These derivations therefore hold without change, and so do their results in operator form: the projection R [2, (1.38), (1.44)], the operator H_k [2, (1.60), (1.103)], the action Δ_k [2, (1.65)], the operator G [2, (1.81), (1.82)], and the identities [2, (1.95), (1.97), (1.99), (1.107)].

The momentum representation, (1.29)–(1.37) and (1.45). The Fourier transform [2, (1.29)] is that of the finite group T_η/Γ', whose dual consists of the classes modulo $2\pi\eta^{-1}\rho\mathbb{Z}^4$ of the p with $p \cdot \gamma \in 2\pi\mathbb{Z}$ for $\gamma \in \Gamma'$, and the decomposition $p = p' + l$ is as described after Lemma 7.1. The symbols [2, (1.31)] are replaced by (7.1) and (7.4). The solution [2, (1.32)–(1.35), (1.37)] uses of u_k only that $u_k(l) = 0$ for the classes $l \neq 0$ and $u_k(0) = 1$, which is Lemma 7.1(iv), and it holds without change. The bound $\Delta_0(p')/\Delta(p' + l) \leq O(1)(1 + |l|^2)^{-1}$ holds with l replaced by $\hat{l} \coloneqq \hat{p} - p'$ for the representative $\hat{p}$ of the proof of Lemma 7.3: by the estimates given there the quotient is at most $\pi^4/|\hat{p}|^2$, and $|\hat{p}| \geq \max(\pi, |\hat{l}| - 2\pi)$ for $\hat{l} \neq 0$. The bound [2, (1.36)] is replaced by (7.7). The formula [2, (1.45)] and the bounds on it are taken from [1], which we have not examined.

The momentum representation, (1.61)–(1.63), (1.66), (1.67), (1.83)–(1.88), and (1.101). The representation [2, (1.61)] of Q_k is replaced by Lemma 7.1(ii): the scalar v_μ becomes the matrix V. The function ϕ_μ of [2, (1.62)] becomes the matrix (7.9), and the bounds

$\gamma_0 \le \Delta_0 \phi_\mu \le \gamma_1$ stated there are Lemma 7.3. The formulae [2, (1.63), (1.66), (1.83)–(1.88), (1.101)] are the operator formulae [2, (1.60), (1.65), (1.81), (1.99)] written in components under the assumption that ϕ is diagonal. In the tilted run they have to be written with the matrices V and ϕ on the indices $\{2,3\}$, in which products such as $\overline{v_\mu}\phi_\mu^{-1}$ become $V^*\phi^{-1}$ and the function ϕ_μ of [2, (1.84)] becomes $\mathbf{1} + a\phi$. Bałaban reads three bounds off from these formulae. Two of them, the inequality [2, (1.67)] and Proposition 1.1 [2, (1.89), (1.90)], are Proposition 9.3, which we have proved without the formulae in components. The third, the bound on the sum over l stated after [2, (1.63)], from which the Hölder continuity of the derivatives of $H_k B$ follows, has not been carried out in this form; by [2, (1.103)], $H_k = GQ_k^*(Q_k G Q_k^*)^{-1}$, the properties of H_k reduce to those of G, which are Proposition 16.2, and to the decay of the kernel of $(Q_k G Q_k^*)^{-1}$, which we have not written out.

Section F, Proposition 1.2. The bounds [2, (1.110)–(1.113)] are Proposition 16.2, and the bounds [2, (1.114)] are Lemma 16.1 as far as the first derivatives are concerned. The proof is organised differently from Bałaban's. His expansion [2, (1.119)–(1.123)] of G over a partition of unity, with the commutators [2, (1.120), (1.121)] of the averages with functions, is not used; the decay comes from a weighted form of the lower bound (9.2). The bounds [2, (1.126), (1.127)] on the kernel of $\partial P \partial^*$ are Lemma 15.1, the identities [2, (1.132), (1.133)] are used as they stand, and the bounds [2, (1.135)–(1.137)] on the scalar operator are Propositions 14.3 and 14.4, which rest on Sections 12 and 13. The expansion over the partition of unity is, however, the model of the expansions of the later papers, and its commutators have not been written for the staircases.

11 The Modifications in the Paper on the Scalar Green's Functions

The paper [1] treats the operators $G_k(\Omega, A) = (-\Delta^\eta_{A,\Omega} + m^2 + aP_k(A))^{-1}$ for functions with values in $\mathbb{R}^N$, on a union Ω of big blocks, with Neumann conditions on its boundary and with an external field A. We have read its Sections 1 and 2, with the following result for the tilted run.

The setting, (1.1)–(1.8). The blocks [1, (1.1)] are those of Definition 6.1(i), and the big blocks are cubes of the unit lattice as before. The quadratic form [1, (1.3)] is the sum over the bonds of T_η with both ends in Ω. The covariant average [1, (1.4)] needs contours inside the blocks, which are those of Definition 8.2.

The random walk representation, (2.2)–(2.13). The partition of unity consists of functions of the coordinates x_μ/M, whose differences along a bond of T_η are $O(\eta/M)$ as before. The formulae [1, (2.3)–(2.5)] concern one bond or the bonds at one point and do not involve the blocks; the commutator [1, (2.8)] of $P_k(A)$ with a function uses the contours of the block only. The derivation of [1, (2.9)–(2.13)] is therefore unchanged.

Lemma 2.1 and Corollary 2.3. The proof of the lemma reduces the L^2-bounds, by the expansion [1, (2.24)–(2.26)] in the regular part of the field, to the lower bound [1, (2.27)] for one unit block at $A = 0$, where the Laplace operator of the block with Neumann conditions is bounded below by π^2 on the functions orthogonal to the constants. In the tilted run this bound is Lemma 8.3, with C_P^{-1} in place of π^2, so that [1, (2.28)] holds with $c_0^{-1} = \min\{C_P^{-1}, a_k\}$, and [1, (2.29)] is unchanged. We have not checked the expansion [1, (2.24)] itself, which is quoted from another paper. Corollary 2.3 follows from the lemma and the random walk representation as in the text.

Lemma 2.2 and the last assertion of the Theorem. These concern a rectangular parallelepiped $\square$ with Neumann conditions, and their proof passes from the operator of the whole

lattice to that of $\square$ by multiple reflections [1, (2.42)]. The reflections in the faces of $\square$ which are orthogonal to e_2 or e_3 do not map T_η to itself: by Lemma 5.1 no reflection of the plane of these two directions preserves both lattices. The formula [1, (2.42)] is therefore not available in the tilted run, and neither are, by this proof, the bounds of Lemma 2.2 in the supremum, Hölder, and L^p norms up to the boundary of a parallelepiped. In the proof of the Theorem the lemma is applied to the cubes $\square_j$ in the interior of Ω, on which the functions h_j vanish near the boundary; a replacement of the operators of these cubes by operators without a boundary has not been carried out.

Lemma 2.4. For the whole lattice the representation [1, (2.43)–(2.48)] is the solution of a linear equation in the momentum representation, and it holds with the function u_j of (7.4). The bounds [1, (2.35), (2.36)], with the analytic continuation in p' on which the exponential decay rests, are Proposition 12.2; of the lower bound on u_j stated in [1, p. 585] for all l, which we have not proved for the tilted u_j, only the case $l = 0$ is used, and this case is (7.8). The bound [1, (2.37)] is Lemma 14.2, and the identity [1, (2.34)], which is quoted there from a paper that is not among our sources, is Lemma 13.1. The resulting bounds for the whole lattice at $A = 0$ are Proposition 14.3.

Sections 3 to 5. Not examined.

12 The Kernel of the Scalar Operator on the Whole Lattice

We now carry out, for the tilted run, the part of the proof of [1, Lemma 2.4] which concerns the whole lattice: the bounds [1, (2.35), (2.36)] on the kernel of $G'Q_k'^*$, with their exponential decay. Bałaban obtains the decay by moving the integration over p' into the complex domain, and says of the required bounds on the integrand in a complex neighbourhood that they are "more troublesome, but equally elementary" [1, p. 586]. For the tilted symbols they follow from two observations: u_k is the Fourier transform of a probability measure on the unit cube, which bounds it and its differences in every strip, and its decay in l is read off from the product (7.4) for complex arguments as for real ones.

In this section the fine lattice is the infinite lattice T_η, $a > 0$, and $G' \coloneqq (-\Delta + aQ_k'^*Q_k')^{-1}$ on the square-summable functions on T_η, where $-\Delta = \partial^*\partial$ and the adjoint is taken with the weights η^4 on T_η and 1 on $\mathbb{Z}^4$, so that $(Q_k'^* g)(x) = g(y)$ for $x \in B_1(y)$ and $Q_k'^*Q_k'$ is the orthogonal projection onto the functions which are constant on the blocks. The inverse exists: by Lemma 8.3 with $a = 1$, applied on every block to the difference of a function and its mean, $-\Delta + aQ_k'^*Q_k' \geq \min(C_P^{-1}, a)$. We write $\mathcal{K}(x, y)$, $x \in T_\eta$, $y \in \mathbb{Z}^4$, for the kernel of $G'Q_k'^*$, that is, $(G'Q_k'^* g)(x) = \sum_y \mathcal{K}(x, y)g(y)$.

The symbols extend to entire functions of $p \in \mathbb{C}^4$: $u_k(p) = \sum_{x \in B_1(0)} \eta^4 e^{ip\cdot x}$, and

$$\Delta(p) \coloneqq \eta^{-2} \sum_\nu (2 - 2\cos(\eta p \cdot e_\nu')) \,, \tag{12.1}$$

which for real p is the function of (7.1); for real p, $\overline{u_k(p)} = u_k(-p)$. Both are periodic with respect to $2\pi\eta^{-1}\rho\mathbb{Z}^4$. For $p \in \mathbb{C}^4$ we define q from the real part of p as before Lemma 7.2, and $\hat{p} \in \mathbb{R}^4$ is the representative of the real part of p modulo $2\pi\eta^{-1}\rho\mathbb{Z}^4$ with $|\eta\hat{p} \cdot e_\nu'| \leq \pi$ for every ν.

Lemma 12.1 (The Symbols in a Strip)**.** *Let $10\eta \leq 1$, and let $p \in \mathbb{C}^4$ with the imaginary part τ, $|\tau_\mu| \leq \frac{1}{4}$ for every μ.*

(i) $|u_k(p)| \leq e^{9/4} \prod_\mu \min(1, \pi/|q_\mu|)$.

(ii) If s is the real part of p, then $|u_k(p) - u_k(s)| \le 2\sum_\mu |\tau_\mu|$.

(iii) $\operatorname{Re}\Delta(p) \ge (4/\pi^2)|\hat p|^2 - 2|\tau|^2$ *and* $|\Delta(p)| \le |\hat p|^2 + 2|\tau|^2$*; and if the real part s of p satisfies* $|\eta s \cdot e'_\nu| \le \pi$ *for every ν, then* $|\Delta(p) - \Delta(s)| \le 2|\tau|^2 + 4|s||\tau|$.

Proof. (i) Let $t := \frac14$. By (7.2), $|D^{a,1}_\mu(p)| \le a\sum_{n<1/a} e^{an|\tau_\mu|} \le e^t$. Further $|e^{ip_\mu} - 1| \le 1 + e^t \le 2e^t$, and for $z = ap_\mu$ with the real part ς and the imaginary part ϑ, $|e^{iz} - 1|^2 = (1 - e^{-\vartheta})^2 + 4e^{-\vartheta}\sin^2(\varsigma/2) \ge e^{-t}(4/\pi^2)(aq_\mu)^2$, because $\varsigma \equiv aq_\mu$ modulo 2π and $|aq_\mu| \le \pi$. Hence $|D^{a,1}_\mu(p)| = |e^{ip_\mu} - 1|a/|e^{iap_\mu} - 1| \le 2e^t e^{t/2}\pi/(2|q_\mu|)$, and altogether $|D^{a,1}_\mu(p)| \le e^{2t}\min(1, \pi/|q_\mu|)$, for $a = \eta$ and for $a = 5\eta$. Since $|c| < 5\sqrt2$ for $c \in \mathsf{C}$ and the imaginary part of $\eta(p_2, p_3)$ has the length at most $\sqrt2\,\eta t$, $|\kappa(\eta p_2, \eta p_3)| \le e^{10\eta t} \le e^t$. The product (7.4) of four factors D and of κ gives (i).

(ii) For $x \in B_1(0)$ we have $0 \le x_\mu < 1$, so that $|\tau \cdot x| \le \sum_\mu |\tau_\mu| \le 1$ and $|e^{ip\cdot x} - e^{is\cdot x}| = |e^{-\tau\cdot x} - 1| \le 2\sum_\mu |\tau_\mu|$, because $e^v - 1 \le 2v$ for $0 \le v \le 1$; and u_k is an average over these x.

(iii) By periodicity we may replace the real part of p by $\hat p$. Let $\varsigma := \eta\hat p \cdot e'_\nu$ and $\vartheta := \eta\tau \cdot e'_\nu$, so that $|\varsigma| \le \pi$ and $|\vartheta| \le \eta|\tau| \le \frac{1}{20}$. Then $2 - 2\cos(\varsigma + i\vartheta) = 2(1 - \cos\varsigma) - 2\cos\varsigma(\cosh\vartheta - 1) + 2i\sin\varsigma\sinh\vartheta$, where $(2/\pi^2)\varsigma^2 \le 1 - \cos\varsigma \le \frac12\varsigma^2$, $\cosh\vartheta - 1 \le \vartheta^2$, and $|\sinh\vartheta| \le 2|\vartheta|$. The real part is therefore at least $(4/\pi^2)\varsigma^2 - 2\vartheta^2$, the modulus is $4\sin^2(\varsigma/2) + 4\sinh^2(\vartheta/2) \le \varsigma^2 + 2\vartheta^2$, and the difference from the value at $\vartheta = 0$ is at most $2\vartheta^2 + 4|\varsigma||\vartheta|$ in modulus. Summing over ν, with $\sum_\nu \varsigma^2 = \eta^2|\hat p|^2$, $\sum_\nu \vartheta^2 = \eta^2|\tau|^2$, and $\sum_\nu |\varsigma||\vartheta| \le \eta^2|\hat p||\tau|$, and dividing by η^2 gives (iii). *q.e.d.*

Proposition 12.2 (The Kernel of $G'Q'^*_k$)**.** *Let $32\eta \le \frac12$ and $0 < a_- \le a \le a_+$. There are constants $\delta_0 > 0$ and c_0, and for every $\alpha \in [0, 1)$ a constant c_α, which depend on a_- and a_+ and not on k and L, such that for all $x, x' \in T_\eta$ with $|x - x'| \le 1$, all $y \in \mathbb{Z}^4$, and all ν,*

$$\begin{aligned} |\mathcal{K}(x,y)|\ ,\ |(\partial'_\nu\mathcal{K})(x,y)| &\le c_0 e^{-\delta_0|x-y|}\ , \\ |(\partial'_\nu\mathcal{K})(x,y) - (\partial'_\nu\mathcal{K})(x',y)| &\le c_\alpha|x-x'|^\alpha e^{-\delta_0\min(|x-y|,|x'-y|)}\ , \end{aligned} \tag{12.2}$$

where ∂'_ν acts on the first argument.

Proof. The representation. We write a momentum of the fine lattice as $p' + l$ with $p' \in [-\pi, \pi]^4$ and l in the set $\mathfrak{L}$ of the η^{-4} classes of $2\pi\mathbb{Z}^4$ modulo $2\pi\eta^{-1}\rho\mathbb{Z}^4$. By Lemma 7.1(i), for f on T_η and g on $\mathbb{Z}^4$, $(\widetilde{Q'_k f})(p') = \sum_l u_k(p' + l)\tilde f(p' + l)$ and $(\widetilde{Q'^*_k g})(p' + l) = u_k(-p' - l)\tilde g(p')$. The equation $(-\Delta + aQ'^*_k Q'_k)\phi = Q'^*_k g$ therefore reads $\Delta(p' + l)\tilde\phi(p' + l) + au_k(-p' - l)w(p') = u_k(-p' - l)\tilde g(p')$ with $w(p') := \sum_l u_k(p' + l)\tilde\phi(p' + l)$; multiplying by $u_k(p' + l)/\Delta(p' + l)$ and summing over l gives $w = \mathsf{T}\tilde g/(1 + a\mathsf{T})$ with $\mathsf{T}(p') = \sum_l u_k(p' + l)u_k(-p' - l)/\Delta(p' + l)$, and then, as in [1, (2.47), (2.48)],

$$\mathcal{K}(x,y) = (2\pi)^{-4}\int_{[-\pi,\pi]^4} \Phi(p')\,\mathrm{d}p'\ , \qquad \Phi(p') := \sum_{l\in\mathfrak{L}} \frac{e^{i(p'+l)\cdot(x-y)}\,u_k(-p'-l)}{\Delta(p'+l)(1 + a\mathsf{T}(p'))}\ . \tag{12.3}$$

The function Φ is periodic with respect to $2\pi\mathbb{Z}^4$, a translation of p' by an element of $2\pi\mathbb{Z}^4$ permuting the classes. We single out the class $l = 0$ and put, for p' in the set U of the $p' \in \mathbb{C}^4$ whose real parts have the coordinates at most $\pi + \frac14$ and whose imaginary parts have the coordinates at most $\frac14$ in modulus,

$$\mathsf{T}_*(p') := \sum_{l\ne0} \frac{u_k(p' + l)u_k(-p' - l)}{\Delta(p' + l)}\ , \qquad \mathcal{D}(p') := \Delta(p')(1 + a\mathsf{T}_*(p')) + au_k(p')u_k(-p')\ , \tag{12.4}$$

so that $\Delta(p')(1+a\mathsf{T}(p')) = \mathcal{D}(p')$ and

$$\Phi(p') = \frac{1}{\mathcal{D}(p')}\Big(e^{ip'\cdot(x-y)}u_k(-p') + \sum_{l\neq 0} e^{i(p'+l)\cdot(x-y)}u_k(-p'-l)\frac{\Delta(p')}{\Delta(p'+l)}\Big)\ . \tag{12.5}$$

The terms $l \neq 0$ on U. Let $p' \in U$ and $l \neq 0$, and let $\hat{p}$ be the representative of the real part of $p'+l$. The difference of $\hat{p}$ and the real part of p' is a nonzero element of $2\pi\mathbb{Z}^4$, by Lemma 7.1(iv), so that one coordinate of $\hat{p}$ is at least $2\pi - \pi - \frac{1}{4}$ in modulus, $|\hat{p}| \geq 2.8$, and by Lemma 12.1(iii), with $|\tau|^2 \leq \frac{1}{4}$,

$$\operatorname{Re}\Delta(p'+l) \geq \frac{4}{\pi^2}|\hat{p}|^2 - \frac{1}{2} \geq \frac{2}{\pi^2}|\hat{p}|^2\ . \tag{12.6}$$

In particular T_* and $\mathcal{D}$ are analytic on U. We need three sums over the classes $l \neq 0$, for $\beta \in \{0, 1, 1+\alpha\}$:

$$\Sigma_\beta(p') := \sum_{l\neq 0} |\hat{p}|^{\beta-2} \prod_\mu \min(1, \pi/|q_\mu|) \leq C_\beta\ , \tag{12.7}$$

with constants C_0, C_1, and $C_{1+\alpha}$ which depend on nothing, respectively on α. To see this, recall that q is of the form $q_\mu = r_\mu + 2\pi n_\mu$ with r the real part of p' and integers n_μ, and that every such q belongs to 25 classes, one of them central. For the central class $\hat{p} = q$, the class is not zero only if $n \neq 0$, and then $|q_\mu| \geq 2\pi|n_\mu| - \pi - \frac{1}{4} \geq |n_\mu|$ and $|\hat{p}| \geq \max_\mu |n_\mu|$, so that the central classes contribute at most $\pi^4 \sum_{n\neq 0} (\max_\mu |n_\mu|)^{\beta-2} \prod_\mu \max(1, |n_\mu|)^{-1} \leq \pi^4 (\sum_{n\in\mathbb{Z}} \max(1,|n|)^{-1-(2-\beta)/4})^4$, which is finite for $\beta < 2$. For the other 24 classes the plane part of $\hat{p}$ is $(q_2, q_3) + \frac{2\pi}{5\eta}m$ with a nonzero $m \in \mathbb{Z}[i]$, and $|(q_2,q_3)| \leq \sqrt{2}\,\pi/(5\eta)$, so that $|\hat{p}| \geq (2-\sqrt{2})\pi/(5\eta) \geq 1/(3\eta)$; the number of the values of n_μ with $|q_\mu| \leq \pi/\eta$ is at most $1/\eta + 2$, and since $\pi/|q_\mu| \leq 2/|n_\mu|$ for $n_\mu \neq 0$, $\sum_{n_\mu} \min(1, \pi/|q_\mu|) \leq 5 + 4\log(2/\eta)$. These classes therefore contribute at most $24(3\eta)^{2-\beta}(5+4\log(2/\eta))^4$, which is bounded on $0 < \eta \leq 1$ for every $\beta < 2$. With (12.6), Lemma 12.1(i), and $\min(1, \pi/|q_\mu|)^2 \leq \min(1, \pi/|q_\mu|)$,

$$|\mathsf{T}_*(p')| \leq \frac{\pi^2}{2} e^{9/2}\, \Sigma_0(p') \leq C_* := \frac{\pi^2}{2} e^{9/2} C_0 \tag{12.8}$$

for every $p' \in U$.

The denominator. Let $s \in [-\pi, \pi]^4$. Then $\Delta(s) \geq 0$, $\mathsf{T}_*(s) \geq 0$, and $u_k(s)u_k(-s) = |u_k(s)|^2 \geq c_u := \frac{1}{4}(2/\pi)^8$ by (7.8), so that $\mathcal{D}(s) \geq ac_u$. Let $\tau \in \mathbb{R}^4$ with $|\tau_\mu| \leq \delta \leq \frac{1}{8}$. The function $z \mapsto \mathsf{T}_*(s + z\tau/\delta)$ is analytic and bounded by C_* on the disc $|z| < \frac{1}{4}$, so that its derivative is at most $8C_*$ on the disc $|z| \leq \frac{1}{8}$, and $|\mathsf{T}_*(s+i\tau) - \mathsf{T}_*(s)| \leq 8C_*\delta$. By Lemma 12.1(ii) and (i), $|u_k(s+i\tau)u_k(-s-i\tau) - |u_k(s)|^2| \leq 8\delta(e^{9/4}+1)$, and by Lemma 12.1(iii), with $|s| \leq 2\pi$ and $|\tau| \leq 2\delta$, $|\Delta(s+i\tau) - \Delta(s)| \leq 8\delta^2 + 16\pi\delta \leq 52\delta$ and $\Delta(s) \leq 4\pi^2$. Hence

$$|\mathcal{D}(s+i\tau) - \mathcal{D}(s)| \leq \delta(52(1+aC_*) + 32\pi^2 aC_* + 84a)\ , \tag{12.9}$$

and we let $\delta_0 \leq \frac{1}{8}$ be so small that the right side with $\delta = \delta_0$ and $a = a_+$ is at most $\frac{1}{2}a_- c_u$. Then $|\mathcal{D}(s+i\tau)| \geq \frac{1}{2}ac_u$ whenever $|\tau_\mu| \leq \delta_0$, the function Φ is analytic in a neighbourhood of the set of these $s+i\tau$, by (12.5), and by its periodicity and Cauchy's theorem in each variable, $\int \Phi(s)\,\mathrm{d}s = \int \Phi(s+i\tau)\,\mathrm{d}s$ over $[-\pi,\pi]^4$.

The bounds. Let $\tau_\mu := \delta_0$ if $x_\mu \geq y_\mu$ and $\tau_\mu := -\delta_0$ otherwise, so that $|e^{i(s+i\tau+l)\cdot(x-y)}| = e^{-\delta_0 \sum_\mu |x_\mu - y_\mu|} \leq e^{-\delta_0|x-y|}$. For $p' = s+i\tau$ we have $|u_k(-p')| \leq e^{9/4}$, $|\Delta(p')| \leq 4\pi^2+1$, and, by (12.6) and Lemma 12.1(i), the sum over $l \neq 0$ in (12.5) is at most $(4\pi^2+1)\frac{\pi^2}{2}e^{9/4}\Sigma_0 e^{-\delta_0|x-y|}$. With (12.7) and the lower bound on $\mathcal{D}$ this is the first bound of (12.2). The difference operator

∂'_ν multiplies the term l by $\eta^{-1}(e^{i\eta(p'+l)\cdot e'_\nu}-1)$, which is at most $2(|\hat p|+1)$ in modulus because $|e^{iz}-1|\le|z|e^{|\mathrm{Im}\,z|}$; the term $l=0$ stays bounded, and the terms $l\neq0$ are bounded through Σ_1, which gives the second bound. For the third we may assume $|x'-y|\le|x-y|$ and choose τ for the pair (x',y). The difference of the two kernels has the terms of $(\partial'_\nu\mathcal{K})(x',y)$ multiplied by $e^{i(p'+l)\cdot(x-x')}-1$, and for $|x-x'|\le1$ this factor is at most $2\min(2,(|\hat p|+1)|x-x'|)\le 4(|\hat p|+1)^\alpha|x-x'|^\alpha$ in modulus. The terms $l\neq0$ are then bounded through $\Sigma_{1+\alpha}$, with a constant which depends on α. *q.e.d.*

The proposition replaces [1, (2.35), (2.36)] for the whole lattice. On a torus T_η/Γ' the kernel is the sum of $\mathcal{K}(x,y+\gamma)$ over $\gamma\in\Gamma'$, which converges by (12.2).

13 The Equation of the Scales

The bounds on the scalar operator in the supremum and Hölder norms are obtained in [1] from the identity (2.34) there, which expresses the operator of k steps through those of fewer steps and through operators on the intermediate lattices, and which is quoted from a paper that is not among our sources. We derive it for the tilted run. Let $\ell_j\coloneqq L^{j-k}$, $1\le j\le k$, be the spacings of the lattices of the run in the unit of the last one, let $a>0$, and put

$$\frac{1}{\alpha_j}\coloneqq\frac{1}{a}\sum_{i=1}^{j}\ell_i^2\,,\qquad P_{(j)}\coloneqq Q'^*_{\ell_j}Q'_{\ell_j}\,,\qquad G_{(j)}\coloneqq(-\Delta+\alpha_jP_{(j)})^{-1}\,,\tag{13.1}$$

on the functions on T_η, or on a torus T_η/Γ'. The adjoints are taken with the weights η^4 on T_η and ℓ_j^4 on $\ell_j\mathbb{Z}^4$, so that $Q'_{\ell_j}Q'^*_{\ell_j}=I$ and $P_{(j)}$ is the orthogonal projection onto the functions which are constant on the blocks $B_{\ell_j}(y)$; the inverse exists by Lemma 8.3 for the scale ℓ_j, as in Section 12. In the unit of the lattice $\ell_j\mathbb{Z}^4$ the constant is $a_j\coloneqq\alpha_j\ell_j^2$, with $1/a_j=a^{-1}\sum_{i=0}^{j-1}L^{-2i}$, so that $a(1-L^{-2})\le a_j\le a$.

Lemma 13.1 (The Equation of the Scales)**.** *For $1\le j\le k-1$ let $Q'^{\mathrm{st}}\coloneqq Q'^{\mathrm{st}}_{\ell_j,\ell_{j+1}}$ be Bałaban's average* (6.4) *from $\ell_j\mathbb{Z}^4$ to $\ell_{j+1}\mathbb{Z}^4$, and let, on the functions on $\ell_j\mathbb{Z}^4$,*

$$\Delta^{(j)}\coloneqq\alpha_j-\alpha_j^2\,Q'_{\ell_j}G_{(j)}Q'^*_{\ell_j}\,,\qquad C^{(j)}\coloneqq(\Delta^{(j)}+a\ell_{j+1}^{-2}Q'^{\mathrm{st}*}Q'^{\mathrm{st}})^{-1}\,.\tag{13.2}$$

Then $\Delta^{(j)}\ge0$, the inverse $C^{(j)}$ exists, and

$$G_{(j+1)}=G_{(j)}+\alpha_j^2\,G_{(j)}Q'^*_{\ell_j}C^{(j)}Q'_{\ell_j}G_{(j)}\,,\qquad G_{(k)}=G_{(1)}+\sum_{j=1}^{k-1}\alpha_j^2\,G_{(j)}Q'^*_{\ell_j}C^{(j)}Q'_{\ell_j}G_{(j)}\,.\tag{13.3}$$

Proof. Write $\mathsf{q}\coloneqq Q'_{\ell_j}$, $\Pi\coloneqq Q'^{\mathrm{st}*}Q'^{\mathrm{st}}$, and $\beta\coloneqq a\ell_{j+1}^{-2}$. By Lemma 6.3, $Q'_{\ell_{j+1}}=Q'^{\mathrm{st}}\mathsf{q}$, so that $P_{(j+1)}=\mathsf{q}^*\Pi\mathsf{q}$, and Π is an orthogonal projection. By (13.1), $1/\alpha_{j+1}=1/\alpha_j+1/\beta$, that is, $\alpha_{j+1}=\alpha_j\beta/(\alpha_j+\beta)$. The operator $\alpha_j-\alpha_j^2(\alpha_j+\beta\Pi)^{-1}$ vanishes on the null space of Π and is $\alpha_j-\alpha_j^2/(\alpha_j+\beta)=\alpha_{j+1}$ on its range, so that it is $\alpha_{j+1}\Pi$, and

$$G_{(j+1)}^{-1}=-\Delta+\alpha_{j+1}\mathsf{q}^*\Pi\mathsf{q}=G_{(j)}^{-1}-\alpha_j^2\,\mathsf{q}^*(\alpha_j+\beta\Pi)^{-1}\mathsf{q}\,.\tag{13.4}$$

For bounded operators M, K with bounded inverses and a bounded operator U such that $\mathsf{M}-\mathsf{U}\mathsf{K}\mathsf{U}^*$ has a bounded inverse, $\mathsf{K}^{-1}-\mathsf{U}^*\mathsf{M}^{-1}\mathsf{U}$ has the bounded inverse $\mathsf{K}+\mathsf{K}\mathsf{U}^*(\mathsf{M}-\mathsf{U}\mathsf{K}\mathsf{U}^*)^{-1}\mathsf{U}\mathsf{K}$, and $(\mathsf{M}-\mathsf{U}\mathsf{K}\mathsf{U}^*)^{-1}=\mathsf{M}^{-1}+\mathsf{M}^{-1}\mathsf{U}(\mathsf{K}^{-1}-\mathsf{U}^*\mathsf{M}^{-1}\mathsf{U})^{-1}\mathsf{U}^*\mathsf{M}^{-1}$, as one checks by multiplication. We apply this with $\mathsf{M}=G_{(j)}^{-1}$, $\mathsf{U}=\alpha_j\mathsf{q}^*$, and $\mathsf{K}=(\alpha_j+\beta\Pi)^{-1}$, for which

$\mathsf{M} - \mathsf{U}\mathsf{K}\mathsf{U}^* = G_{(j+1)}^{-1}$ by (13.4) and $\mathsf{K}^{-1} - \mathsf{U}^*\mathsf{M}^{-1}\mathsf{U} = \alpha_j + \beta\Pi - \alpha_j^2 \mathsf{q} G_{(j)} \mathsf{q}^* = \Delta^{(j)} + \beta\Pi$. This gives the existence of $C^{(j)}$ and the first identity of (13.3), and the second follows by adding the first over j. Finally $G_{(j)}^{-1} \geq \alpha_j \mathsf{q}^*\mathsf{q}$ gives $\langle \mathsf{q}^* w, G_{(j)} \mathsf{q}^* w \rangle = \sup_\phi (2\langle w, \mathsf{q}\phi\rangle - \langle \phi, G_{(j)}^{-1}\phi\rangle) \leq \sup_\phi (2\langle w, \mathsf{q}\phi\rangle - \alpha_j \|\mathsf{q}\phi\|^2) \leq \|w\|^2/\alpha_j$, hence $\Delta^{(j)} \geq 0$. *q.e.d.*

The second identity of (13.3) is [1, (2.34)] for the whole lattice or a torus, with $G_{(1)}$ for the operator $C^{(0),\eta}$ there; the factor $(L^j\eta)^{-4}$ of that formula is contained in our adjoints. The operators $G_{(j)}$, Q'_{ℓ_j} are those of the tilted run at the scale ℓ_j, to which Proposition 12.2 applies after a change of the unit of length, and the operators $C^{(j)}$ act on the lattices $\ell_j\mathbb{Z}^4$ of the paper.

14 The Scalar Operator in the Supremum and Hölder Norms

With the equation of the scales and the kernel of Section 12 we obtain the bounds of [1, (1.9), (1.10)] for the whole lattice at the field $A = 0$, by the summation [1, (2.38), (2.39)]. This replaces Lemma 2.2 of that paper, which concerns a box, by a statement without a boundary; the lattice in this section is T_η, and the statements hold on a torus T_η/Γ' with the kernels summed over the periods. Two further inputs are needed, the decay of the kernels of $G_{(1)}$ and of $C^{(j)}$. Bałaban derives the second from his Proposition 2.3; we use that both are inverses of operators which are bounded below and have kernels of rapid decay.

Lemma 14.1 (Inverses of Operators with Decaying Kernels)**.** *Let Y be a countable set with a metric d, and let M be a bounded self-adjoint operator on the square-summable functions on Y, with the kernel $\mathsf{M}(y, y')$, such that $\mathsf{M} \geq \gamma > 0$ and, for some $\theta > 0$ and every y, $\sum_{y'} |\mathsf{M}(y, y')|(e^{\theta d(y,y')} - 1) \leq \frac{1}{2}\gamma$. Then $|\mathsf{M}^{-1}(y, y')| \leq (2/\gamma) e^{-\theta d(y,y')}$ for all $y, y' \in Y$.*

Proof. Fix $y_0 \in Y$ and $R > 0$, and let $w(y) := \min(\theta d(y, y_0), R)$, a bounded function with $|w(y) - w(y')| \leq \theta d(y, y')$. The operator $\mathsf{M}_w := e^w \mathsf{M} e^{-w}$ has the kernel $e^{w(y)-w(y')}\mathsf{M}(y, y')$, and the kernel of $\mathsf{M}_w - \mathsf{M}$ is at most $|\mathsf{M}(y, y')|(e^{\theta d(y,y')} - 1)$ in modulus, a bound which is symmetric in y and y'; by the hypothesis and Schur's test $\|\mathsf{M}_w - \mathsf{M}\| \leq \frac{1}{2}\gamma$. Hence $\|\mathsf{M}_w \phi\| \geq \frac{1}{2}\gamma\|\phi\|$ for every ϕ, and the same holds for the adjoint, so that M_w is invertible with $\|\mathsf{M}_w^{-1}\| \leq 2/\gamma$. Since $e^{\pm w}$ are bounded, $\mathsf{M}_w^{-1} = e^w \mathsf{M}^{-1} e^{-w}$, and $|\mathsf{M}^{-1}(y, y_0)| = e^{-w(y)+w(y_0)} |\mathsf{M}_w^{-1}(y, y_0)| \leq (2/\gamma) e^{-w(y)}$. Letting $R \to \infty$ gives the claim. *q.e.d.*

Lemma 14.2 (The Operators of One Scale)**.** *Let $L \geq 64$ and $a > 0$. There are positive constants c_1, θ_1, c_C, θ_C, which depend on a and L and not on k, such that the kernel g_1 of $G_{(1)}$ and the kernels C_j of $C^{(j)}$, $1 \leq j \leq k-1$, with respect to the sums over the points of T_η and of $\ell_j\mathbb{Z}^4$, satisfy*

$$|g_1(x, x')| \leq c_1 \eta^2 e^{-\theta_1 |x-x'|/\eta}\,, \qquad |\mathsf{C}_j(y, y')| \leq c_C \ell_j^2 e^{-\theta_C |y-y'|/\ell_j}\,. \tag{14.1}$$

Proof. For g_1, let $\mathsf{M} := \eta^2(-\Delta + \alpha_1 P_{(1)})$ on the functions on T_η, with the metric $d(x, x') := |x - x'|/\eta$. Since $\eta^2\alpha_1 = a/L^2$, its kernel with respect to the sum over the points is 8 on the diagonal and -1 for the eight neighbours of a point, plus $(a/L^2)L^{-4}$ for two points of one block $B_{\ell_1}(y)$, whose distance d is at most $2L$. By Lemma 8.3 for the scale $\ell_1 = L\eta$, $\mathsf{M} \geq L^{-2}\min(C_P^{-1}, a)$, and $\sum_{x'} |\mathsf{M}(x, x')|(e^{\theta d(x,x')} - 1) \leq 8(e^\theta - 1) + (a/L^2)(e^{2L\theta} - 1)$, which is at most half this lower bound for $\theta = \theta_1$ small. Lemma 14.1 and $G_{(1)} = \eta^2 \mathsf{M}^{-1}$ give the first bound.

For C_j we take ℓ_j as the unit of length. Then T_η becomes the fine lattice of the spacing L^{-j}, $G_{(j)}$ becomes $\ell_j^2 G'$ with the operator G' of Section 12 for the constant $a_j \in [a(1-L^{-2}), a]$, and $C^{(j)} = \ell_j^2 \mathsf{M}^{-1}$ with $\mathsf{M} := a_j - a_j^2 Q' G' Q'^* + (a/L^2)\Pi$ on the functions on $\mathbb{Z}^4$, where Π is the orthogonal projection onto the functions which are constant on the cubes of side L. The kernel of $Q'G'Q'^*$ is the average of $\mathcal{K}(x, y')$ over $x \in B_1(y)$, which by (12.2) is at most $c_0 e^{2\delta_0} e^{-\delta_0 |y-y'|}$ in modulus, and the kernel of Π is L^{-4} for two points of one cube. Hence $\sum_{y'} |\mathsf{M}(y, y')| (e^{\theta|y-y'|} - 1)$ tends to zero with θ, uniformly in k and j. For the lower bound we use the momentum representation: $Q'G'Q'^*$ is the multiplication by $\mathsf{T}(p')/(1 + a_j \mathsf{T}(p'))$, by the proof of Proposition 12.2, so that $a_j - a_j^2 Q'G'Q'^*$ is the multiplication by $a_j/(1 + a_j \mathsf{T}(p'))$, and $\mathsf{T} \le C_\mathsf{T} \Delta_0^{-1}$ by Lemma 9.2, whose proof applies to the whole lattice. With $\Delta_0 \le 16$ this gives $a_j - a_j^2 Q'G'Q'^* \ge c_\Delta \Delta_0$, $c_\Delta := a_j/(16 + a_j C_\mathsf{T})$. The Laplace operator Δ_0 of $\mathbb{Z}^4$ is at least the sum of those of the cubes of side L with Neumann conditions, each of which is at least $2/L^2$ on the functions of mean zero by the inequality for the cube of the corners in the proof of Lemma 8.3. Therefore $\mathsf{M} \ge L^{-2} \min(2c_\Delta, a)$, and Lemma 14.1 gives the second bound after the return to the original unit. *q.e.d.*

Proposition 14.3 (The Scalar Operator on the Whole Lattice)**.** *Let $L \ge 64$, so that $32L^{-j} \le \frac{1}{2}$ for every $j \ge 1$, and let $a > 0$. There are constants $\delta > 0$ and c, and for every $\alpha \in [0, 1)$ a constant c'_α, which depend on a and L and not on k, such that for every bounded function f on T_η, all $x, x' \in T_\eta$ with $|x - x'| \le 1$, and every ν,*

$$\begin{aligned} |(G_{(k)} f)(x)| \,,\ |(\partial'_\nu G_{(k)} f)(x)| &\le c\, e^{-\delta \operatorname{dist}(x, \operatorname{supp} f)} \sup|f| \,, \\ |(\partial'_\nu G_{(k)} f)(x) - (\partial'_\nu G_{(k)} f)(x')| &\le c'_\alpha |x - x'|^\alpha e^{-\delta \operatorname{dist}(\{x, x'\}, \operatorname{supp} f)} \sup|f| \,. \end{aligned} \tag{14.2}$$

Proof. By Lemma 13.1, $G_{(k)} f = G_{(1)} f + \sum_{j=1}^{k-1} \mathsf{T}_j f$ with $\mathsf{T}_j := \alpha_j^2 G_{(j)} Q'^*_{\ell_j} C^{(j)} Q'_{\ell_j} G_{(j)}$. Let $\mathcal{K}_j(x, y)$, $x \in T_\eta$, $y \in \ell_j \mathbb{Z}^4$, be the kernel of $G_{(j)} Q'^*_{\ell_j}$. In the unit ℓ_j it is ℓ_j^2 times the kernel of Proposition 12.2 for the constant a_j and the fine spacing L^{-j}, so that, with the constants of that proposition for $a_- = a(1 - L^{-2})$ and $a_+ = a$,

$$|\mathcal{K}_j(x, y)| \le c_0 \ell_j^2 e^{-\delta_0 |x-y|/\ell_j} \,, \qquad |(\partial'_\nu \mathcal{K}_j)(x, y)| \le c_0 \ell_j e^{-\delta_0 |x-y|/\ell_j} \,, \tag{14.3}$$

and $|(\partial'_\nu \mathcal{K}_j)(x, y) - (\partial'_\nu \mathcal{K}_j)(x', y)| \le c_\alpha \ell_j^{1-\alpha} |x - x'|^\alpha e^{-\delta_0 \min(|x-y|, |x'-y|)/\ell_j}$ for $|x - x'| \le \ell_j$; for $\ell_j < |x - x'| \le 1$ the same bound holds with $2c_0$ in place of c_α, by the second bound of (14.3). With the weights of the adjoints, $(Q'_{\ell_j} G_{(j)} f)(y') = \ell_j^{-4} \sum_{x'} \eta^4 \mathcal{K}_j(x', y') f(x')$, and therefore

$$(\mathsf{T}_j f)(x) = \alpha_j^2 \sum_{y, y'} \mathcal{K}_j(x, y)\, \mathsf{C}_j(y, y')\, \ell_j^{-4} \sum_{x'} \eta^4 \mathcal{K}_j(x', y') f(x') \,. \tag{14.4}$$

The sums $\sum_{x'} \eta^4 e^{-\delta|x'-y'|/\ell_j}$ over T_η and $\sum_y e^{-\delta|y-y'|/\ell_j}$ over $\ell_j \mathbb{Z}^4$ are at most a constant, depending on δ, times ℓ_j^4 and times 1, because $\eta \le \ell_j$. Taking half of each exponent for the summations and the other half for the decay, (14.3), (14.1), and $\alpha_j^2 = a_j^2 \ell_j^{-4}$ give

$$|(\mathsf{T}_j f)(x)| \le c'' \ell_j^{-4} \cdot \ell_j^2 \cdot \ell_j^2 \cdot \ell_j^{-4} \ell_j^4 \ell_j^2\, e^{-\delta \operatorname{dist}(x, \operatorname{supp} f)/\ell_j} \sup|f| = c'' \ell_j^2\, e^{-\delta \operatorname{dist}(x, \operatorname{supp} f)/\ell_j} \sup|f| \tag{14.5}$$

with $\delta := \frac{1}{2} \min(\delta_0, \theta_C)$, and in the same way $|(\partial'_\nu \mathsf{T}_j f)(x)| \le c'' \ell_j e^{-\delta \operatorname{dist}(x, \operatorname{supp} f)/\ell_j} \sup|f|$ and the Hölder bound with the factor $\ell_j^{1-\alpha} |x - x'|^\alpha$. Since $\ell_j = L^{j-k} \le 1$, the sums over j of ℓ_j^2, ℓ_j, and $\ell_j^{1-\alpha}$ are bounded uniformly in k, the last by a constant which depends on α. For the first term, (14.1) gives $|(G_{(1)} f)(x)| \le c\eta^2 e^{-\theta_1 \operatorname{dist}(x, \operatorname{supp} f)/(2\eta)} \sup|f|$, the bound with η in place of η^2 for its difference quotients, and, because two different points of T_η are at least η apart, the Hölder bound with the factor $2\eta^{1-\alpha} |x - x'|^\alpha$. *q.e.d.*

The proposition is [1, (1.9), (1.10)] for the whole lattice and $A = 0$, proved along [1, (2.38), (2.39)]. The constants depend on L, through Lemma 14.2, as Bałaban expects for the operators $C^{(j)}$ [1, p. 588].

The first paper on the propagators also needs the scalar operator applied to a derivative of the source, and, for sources which are Hölder continuous, two derivatives [2, (1.135)–(1.137), pp. 39–40]. We write $(\partial'^{*}_{\nu} f)(x) \coloneqq \eta^{-1}(f(x-\eta e'_\nu)-f(x))$ for the adjoint of ∂'_ν, and for $0 < \epsilon \le 1$ we let $\|f\|_\epsilon$ be the supremum of $|f(x)-f(x')|/|x-x'|^\epsilon$ over the pairs of different points with $|x-x'| \le 1$, as in [2, (1.109)].

Proposition 14.4 (Derivatives of the Source)**.** *Let $L \ge 64$ and $a > 0$, and let δ be as in Proposition 14.3. For $\alpha \in [0,1)$ and $\epsilon > 0$ with $\alpha+\epsilon < 1$ there are constants c, which depend on a, L, α, and ϵ and not on k, such that for every bounded function f on T_η, all $x, x' \in T_\eta$ with $|x-x'| \le 1$, and all ν, ν', with $D \coloneqq \operatorname{dist}(\{x,x'\}, \operatorname{supp} f)$,*

(i) $|(G_{(k)}\partial'^{*}_{\nu} f)(x)| \le c\, e^{-\delta D} \sup|f|$ *and* $|(G_{(k)}\partial'^{*}_{\nu} f)(x) - (G_{(k)}\partial'^{*}_{\nu} f)(x')| \le c|x-x'|^\alpha e^{-\delta D} \sup|f|$*;*

(ii) $|(\partial'_{\nu'} G_{(k)} \partial'^{*}_{\nu} f)(x)| \le c\, e^{-\delta D/2}(\|f\|_\epsilon + \sup|f|)$*;*

(iii) $|(\partial'_{\nu'} G_{(k)} \partial'^{*}_{\nu} f)(x) - (\partial'_{\nu'} G_{(k)} \partial'^{*}_{\nu} f)(x')| \le c|x-x'|^\alpha e^{-\delta D/2}(\|f\|_{\alpha+\epsilon} + \sup|f|)$.

Proof. We use the decomposition of the proof of Proposition 14.3. By a summation by parts, $\sum_{x'} \eta^4 \mathcal{K}_j(x',y')(\partial'^{*}_{\nu} f)(x') = \sum_{x'} \eta^4 (\partial'_\nu \mathcal{K}_j)(x',y') f(x')$, so that $\mathsf{T}_j \partial'^{*}_{\nu} f$ is (14.4) with $\partial'_\nu \mathcal{K}_j$ in place of the last kernel. By (14.3) this replaces one factor ℓ_j^2 of (14.5) by ℓ_j: $|(\mathsf{T}_j \partial'^{*}_{\nu} f)(x)| \le c'' \ell_j e^{-\delta D/\ell_j} \sup|f|$, and the difference of the values at x and x' is at most $c'' \min(2\ell_j, |x-x'|) e^{-\delta D/\ell_j} \sup|f| \le 2c'' \ell_j^{1-\alpha} |x-x'|^\alpha e^{-\delta D/\ell_j} \sup|f|$, by the bound on $\partial' \mathcal{K}_j$. The sums over j of ℓ_j and $\ell_j^{1-\alpha}$ are bounded, which gives (i) for these terms.

For (ii) and (iii), $(\partial'_{\nu'} \mathsf{T}_j \partial'^{*}_{\nu} f)(x) = \sum_{x''} \eta^4 k_j(x,x'') f(x'')$ with the kernel

$$k_j(x,x'') \coloneqq \alpha_j^2 \ell_j^{-4} \sum_{y,y'} (\partial'_{\nu'} \mathcal{K}_j)(x,y)\, \mathsf{C}_j(y,y')\, (\partial'_{\nu} \mathcal{K}_j)(x'',y') \,, \tag{14.6}$$

for which (14.3), its Hölder counterpart, and (14.1) give, as in (14.5),

$$\begin{aligned} |k_j(x,x'')| &\le c'' \ell_j^{-4} e^{-\delta |x-x''|/\ell_j} \,, \\ |k_j(x,x'') - k_j(x',x'')| &\le c'' \ell_j^{-4} \Big(\frac{|x-x'|}{\ell_j}\Big)^\alpha e^{-\delta \min(|x-x''|, |x'-x''|)/\ell_j} \,, \end{aligned} \tag{14.7}$$

the second for $|x-x'| \le \ell_j$. Since ∂'^{*}_{ν} annihilates the constants and the sums converge absolutely, $\sum_{x''} \eta^4 k_j(x,x'') = 0$ for every x. Hence $(\partial'_{\nu'} \mathsf{T}_j \partial'^{*}_{\nu} f)(x) = \sum_{x''} \eta^4 k_j(x,x'')(f(x'') - f(x))$. Here $|f(x'')-f(x)| \le \|f\|_\epsilon |x-x''|^\epsilon$ for $|x-x''| \le 1$ and $|f(x'')-f(x)| \le 2\sup|f|$ otherwise, and $\sum_{x''} \eta^4 \ell_j^{-4} e^{-\delta|x-x''|/\ell_j} |x-x''|^\epsilon$ is at most a constant times ℓ_j^ϵ, as is $e^{-\delta/(2\ell_j)}$. This gives $|(\partial'_{\nu'} \mathsf{T}_j \partial'^{*}_{\nu} f)(x)| \le c\, \ell_j^\epsilon (\|f\|_\epsilon + \sup|f|)$, with a sum over j which is bounded. If $D \ge 1$, then f vanishes at x and the first bound of (14.7) gives directly $|(\partial'_{\nu'} \mathsf{T}_j \partial'^{*}_{\nu} f)(x)| \le c\, e^{-\delta D/(2\ell_j)} \sup|f|$, and $\sum_j e^{-\delta D/(2\ell_j)} \le e^{-\delta D/2} \sum_{i \ge 0} e^{-\delta(L^i-1)/2}$. The two cases give (ii). For (iii), the vanishing of the sums of $k_j(x,\cdot)$ and of $k_j(x',\cdot)$ gives

$$(\partial'_{\nu'} \mathsf{T}_j \partial'^{*}_{\nu} f)(x) - (\partial'_{\nu'} \mathsf{T}_j \partial'^{*}_{\nu} f)(x') = \sum_{x''} \eta^4 (k_j(x,x'') - k_j(x',x''))(f(x'') - f(x)) \,. \tag{14.8}$$

For $|x-x'| \le \ell_j$ the second bound of (14.7) and the estimate of the sum, with $\alpha+\epsilon$ in place of ϵ, bound this by $c(|x-x'|/\ell_j)^\alpha \ell_j^{\alpha+\epsilon}(\|f\|_{\alpha+\epsilon} + \sup|f|)$; for $|x-x'| > \ell_j$ the two terms are

bounded separately by $c\,\ell_j^{\alpha+\epsilon} \le c|x-x'|^\alpha \ell_j^\epsilon$ times the same norms. For $D \ge 1$ the decay is obtained as in (ii), with $\sum_i L^{i\alpha} e^{-\delta(L^i-1)/2} < \infty$.

The term $G_{(1)}$ is treated in the same way with η in place of ℓ_j: by (14.1) the kernels of $G_{(1)}\partial'^*_\nu$ and of $\partial'_{\nu'} G_{(1)} \partial'^*_\nu$ with respect to the weights η^4 are at most a constant times $\eta^{-3} e^{-\theta_1|x-x''|/(2\eta)}$ and $\eta^{-4} e^{-\theta_1|x-x''|/(2\eta)}$, the second has vanishing sums over x'', and two different points are at least η apart. *q.e.d.*

Statement (ii) is [2, (1.137)], and (i) and (iii) are the corresponding bounds for the scalar operator in [2, (1.111), (1.113)].

15 The Kernel of the Gauge-Fixing Projection

The operator Δ_a of [2, (1.69)] contains the nonlocal term $\partial P \partial^*$, and the passage from L^2-bounds to bounds in the supremum norm needs the bounds [2, (1.126), (1.127)] on its kernel. Bałaban refers for them to the representation [2, (1.44)], $P = G' Q'^*_k (Q'_k G'^2 Q'^*_k)^{-1} Q'_k G'$, to [1, Lemma 2.4], and to the analyticity of the momentum representation [2, (1.45)]. In the tilted run the first two are available by Proposition 12.2, and for the inverse in the middle we use Lemma 14.1.

Lemma 15.1 (The Kernel of $\partial P \partial^*$)**.** *Let $32\eta \le \frac12$, let $a > 0$, and let G' and $\mathcal{K}$ be as in Section 12. The operator $\mathsf{N} := Q'_k G'^2 Q'^*_k$ on the square-summable functions on $\mathbb{Z}^4$ has a bounded inverse, $P := G' Q'^*_k \mathsf{N}^{-1} Q'_k G'$ is the orthogonal projection onto the orthogonal complement of $\Delta N(Q'_k)$, and for every field A of finite support*

$$\begin{aligned} (\partial P \partial^* A)_\nu(x) &= \sum_{\nu'} \sum_{x'} \eta^4 \varpi_{\nu\nu'}(x,x') A_{\nu'}(x') \;, \\ \varpi_{\nu\nu'}(x,x') &:= \sum_{y,y'} (\partial'_\nu \mathcal{K})(x,y)\, \mathsf{N}^{-1}(y,y')\, (\partial'_{\nu'} \mathcal{K})(x',y') \;. \end{aligned} \tag{15.1}$$

There are constants c_ϖ and $\delta_\varpi > 0$, and for $\alpha \in [0,1)$ a constant $c_{\varpi,\alpha}$, which depend on a and not on k and L, such that for $|x-x'| \le 1$

$$\begin{aligned} |\varpi_{\nu\nu'}(x,x'')| &\le c_\varpi e^{-\delta_\varpi |x-x''|} \;, \\ |\varpi_{\nu\nu'}(x,x'') - \varpi_{\nu\nu'}(x',x'')| &\le c_{\varpi,\alpha} |x-x'|^\alpha e^{-\delta_\varpi \min(|x-x''|,|x'-x''|)} \;. \end{aligned} \tag{15.2}$$

Proof. With the weights of Section 12, $(Q'_k G' f)(y) = \sum_x \eta^4 \mathcal{K}(x,y) f(x)$, so that N has the kernel $\mathsf{N}(y,y') = \sum_x \eta^4 \mathcal{K}(x,y)\mathcal{K}(x,y')$, which by (12.2) is at most a constant times $e^{-\delta_0|y-y'|/2}$ in modulus. In the momentum representation N is the multiplication by $\sum_l |u_k(p'+l)|^2 \Delta(p'+l)^{-2}(1+a\mathsf{T}(p'))^{-2}$, by the proof of Proposition 12.2, and the term $l=0$ of this sum is $|u_k(p')|^2/\mathcal{D}(p')^2$ with $\mathcal{D}$ of (12.4). For real $p' \in [-\pi,\pi]^4$ we have $\mathcal{D}(p') \le 4\pi^2(1+aC_*)+a$ by (12.8) and $|u_k(p')|^2 \ge c_u$, so that $\mathsf{N} \ge \gamma_\mathsf{N} := c_u(4\pi^2(1+aC_*)+a)^{-2}$. Lemma 14.1 gives $|\mathsf{N}^{-1}(y,y')| \le (2/\gamma_\mathsf{N}) e^{-\theta_\mathsf{N}|y-y'|}$ with a $\theta_\mathsf{N} > 0$ which depends on a only.

The operator P is self-adjoint, and $P^2 = P$ because $Q'_k G' G' Q'^*_k = \mathsf{N}$. Its range is the range of $G' Q'^*_k$, whose orthogonal complement consists of the f with $Q'_k G' f = 0$, that is, of the $f = (-\Delta + a Q'^*_k Q'_k)\lambda = -\Delta\lambda$ with $Q'_k \lambda = 0$. This is the second assertion, and P is the operator of [2, (1.38), (1.44)]. For a field A of finite support, $\partial^* A = \sum_{\nu'} \partial'^*_{\nu'} A_{\nu'}$, and a summation by parts gives $(Q'_k G' \partial^* A)(y') = \sum_{\nu'} \sum_{x'} \eta^4 (\partial'_{\nu'} \mathcal{K})(x',y') A_{\nu'}(x')$; applying N^{-1}, then $G' Q'^*_k$ with its kernel $\mathcal{K}$, and ∂'_ν gives (15.1). The bounds (15.2) follow from (12.2) and the bound on N^{-1} by summation over y and y', with $\delta_\varpi := \frac12 \min(\delta_0, \theta_\mathsf{N})$. *q.e.d.*

The bounds (15.2) are [2, (1.126), (1.127)].

16 Proposition 1.2 for the Tilted Run

We can now prove the bounds of [2, Proposition 1.2] on the operator $G = \Delta_a^{-1}$ in the supremum and Hölder norms, with their exponential decay. Bałaban derives the decay from an expansion of G over a partition of unity [2, (1.119)–(1.131)], whose terms contain the commutators of the averages with functions. We obtain it more directly, from the stability of the lower bound (9.2) under the conjugation with an exponential weight, the method which he mentions first [2, p. 36]; the passage from L^2 to the supremum norm is then his, through the identities [2, (1.132), (1.133)].

In this section the lattice is a torus T_η/Γ' with $\Gamma' \subset \mathbb{Z}^4$, $d(x,x')$ is the distance on it, and $L \geq 64$. The operators of the whole lattice with the kernels of Sections 12–15 act on the bounded functions, and they map periodic functions to periodic functions; since each of them is the inverse, or is built from inverses, of operators which are bounded below on the torus as on the whole lattice, their restrictions to the periodic functions are the corresponding operators of the torus. Propositions 14.3 and 14.4 and Lemma 15.1 therefore apply to the torus, with sums over the points of T_η and with the distance to the periodic support, which is the distance on the torus. For a subset X of the torus, $\|\cdot\|_X$ is the L^2-norm over the points, or the bonds starting, in X, and $\Delta(y) \coloneqq B_1(y)$ is the unit block of $y \in \mathbb{Z}^4$. We write $\|A\|_1^2 \coloneqq \langle A, (\Delta + I)A\rangle = \|\nabla A\|^2 + \|A\|^2$, and for a tuple $J = (J_\nu)$ of fields, $\nabla^* J \coloneqq \sum_\nu \partial_\nu'^* J_\nu$.

Lemma 16.1 (Decay in L^2)**.** *Let $a > 0$. There are constants $\theta_0 > 0$ and C, which depend on a and not on k, L, and the torus, such that for all $y, y' \in \mathbb{Z}^4$ and every field J, or tuple of fields J, supported in the bonds starting in $\{x \,|\, d(x,y') \leq 2\}$,*

$$\begin{aligned} \|GJ\|_{\Delta(y)} + \|\nabla GJ\|_{\Delta(y)} &\leq Ce^{-\theta_0 d(y,y')}\|J\| \,, \\ \|G\nabla^* J\|_{\Delta(y)} + \|\nabla G\nabla^* J\|_{\Delta(y)} &\leq Ce^{-\theta_0 d(y,y')}\|J\| \,. \end{aligned} \tag{16.1}$$

The same holds for $G_0 \coloneqq (\Delta + aQ_k^* Q_k)^{-1}$ in place of G.

Proof. Let w be a real function on the torus with $|w(x) - w(x')| \leq \theta d(x,x')$, $\theta \leq \frac{1}{4}$, acting on a field by multiplication with its value at the starting point of the positive bond. We compare the forms of $e^w \Delta_a e^{-w}$ and of Δ_a, term by term.

For the Laplace operator, $\langle A', e^w \Delta e^{-w} A\rangle = \sum_\nu \langle \partial_\nu'(e^w A'), \partial_\nu'(e^{-w}A)\rangle$. With $\varsigma \coloneqq w(x + \eta e_\nu') - w(x)$, $|\varsigma| \leq \theta\eta$, we have $\partial_\nu'(e^{\pm w} g)(x) = e^{\pm w(x)}(e^{\pm\varsigma}(\partial_\nu' g)(x) + \eta^{-1}(e^{\pm\varsigma} - 1)g(x))$, where $e^{|\varsigma|} \leq 2$ and $\eta^{-1}|e^{\pm\varsigma} - 1| \leq 2\theta$. The product of the two derivatives therefore differs from $(\partial_\nu' A')(\partial_\nu' A)$ by at most $4\theta(|\partial_\nu' A'||A| + |A'||\partial_\nu' A|) + 4\theta^2 |A'||A|$ at every bond, and summing gives $|\langle A', (e^w \Delta e^{-w} - \Delta)A\rangle| \leq 16\theta \|A'\|_1 \|A\|_1$.

For the average, $\langle A', e^w Q_k^* Q_k e^{-w} A\rangle = \sum_c (Q_k e^w A')(c)(Q_k e^{-w} A)(c)$, and for a bond c from y the two factors may be replaced by $Q_k(e^{w - w(y)} A')(c)$ and $Q_k(e^{-(w-w(y))}A)(c)$. The bonds which enter the average at c start at a distance of at most 4 from y, so that $|e^{\pm(w-w(y))} - 1| \leq e^{4\theta} - 1 \leq 8\theta$ on them, and by the estimate in the proof of Lemma 9.1(i), which bounds the square of each average by a sum over its own bonds, $|\langle A', (e^w Q_k^* Q_k e^{-w} - Q_k^* Q_k)A\rangle| \leq (32\theta + 128\theta^2)\|A'\|\|A\| \leq 64\theta \|A'\|\|A\|$.

For the projection, $e^w \partial P \partial^* e^{-w} - \partial P \partial^*$ has the kernel $\varpi(x,x')(e^{w(x) - w(x')} - 1)$ with respect to the sums over the points of T_η with the weights η^4, and $|e^{w(x)-w(x')} - 1| \leq \theta |x - x'| e^{\theta|x-x'|}$.

For $\theta \le \frac{1}{2}\delta_\varpi$, (15.2) and Schur's test bound its norm by $C_\varpi\theta$, with $C_\varpi := 4c_\varpi \sup_x \sum_{x'} \eta^4|x-x'|e^{-\delta_\varpi|x-x'|/2}$, which is finite and bounded uniformly in η.

Together, $|\langle A', (e^w\Delta_a e^{-w} - \Delta_a)A\rangle| \le C_w\theta\|A'\|_1\|A\|_1$ with $C_w := 16 + 64a + C_\varpi$. Let $\theta_0 := \min(\frac{1}{4}, \frac{1}{2}\delta_\varpi, \gamma_0/(2C_w))$ with γ_0 of (9.2), and let $\theta \le \theta_0$. Then $\langle F, e^w\Delta_a e^{-w}F\rangle \ge \frac{1}{2}\gamma_0\|F\|_1^2$ for every F, so that $e^w\Delta_a e^{-w}$ is invertible, and its inverse $G_w = e^wGe^{-w}$ satisfies $\|G_wJ\|_1 \le (2/\gamma_0)\|J\|$ and, because $|\langle F, \nabla^*J\rangle| \le \|F\|_1\|J\|$, also $\|G_w\nabla^*J\|_1 \le (2/\gamma_0)\|J\|$.

Now let $w(x) := \theta_0 \max(0, d(x, y') - 3)$. It vanishes on the points at a distance of at most 3 from y', hence on the supports of J and of ∇^*J, so that $GJ = e^{-w}G_wJ$ and $G\nabla^*J = e^{-w}G_w\nabla^*J$; and $w \ge \theta_0(d(y, y') - 5)$ on $\Delta(y)$. Since $|\partial'_\nu(e^{-w}F)| \le 2e^{-w}(|\partial'_\nu F| + \theta_0|F|)$ by the formula above, this gives (16.1) with $C := 8e^{5\theta_0}/\gamma_0$. For G_0 the term with the projection is absent, and the lower bound is $\Delta + aQ_k^*Q_k \ge \min(a/4, (2C_P + 65)^{-1})(\Delta + I)$, by Lemma 9.1(iv). *q.e.d.*

The lemma is [2, (1.114)] for the first derivatives.

Proposition 16.2 (Proposition 1.2 for the Tilted Run)**.** *Let $L \ge 64$ and $a > 0$. For $\alpha \in [0, 1)$ and $\epsilon > 0$ with $\alpha + \epsilon < 1$ there are constants c and $\delta_G > 0$, which depend on a, L, α, and ϵ and not on k and the torus, such that for every bounded field J, or tuple of fields J, all points x, x' with $d(x, x') \le 1$, and $D := d(\{x, x'\}, \operatorname{supp} J)$,*

(i) $|(GJ)(x)|$, $|(\nabla GJ)(x)|$, $|(G\nabla^*J)(x)|$, *and* $|(\Delta GJ)(x)|$ *are at most* $c\,e^{-\delta_G D}\sup|J|$*;*

(ii) the differences of the values of ∇GJ, *and of* $G\nabla^*J$, *at* x *and* x' *are at most* $c\,d(x, x')^\alpha e^{-\delta_G D}\sup|J|$ *in modulus;*

(iii) $|(\nabla G\nabla^*J)(x)| \le c\,e^{-\delta_G D}(\|J\|_\epsilon + \sup|J|)$*;*

(iv) $|(\nabla G\nabla^*J)(x) - (\nabla G\nabla^*J)(x')| \le c\,d(x, x')^\alpha e^{-\delta_G D}(\|J\|_{\alpha+\epsilon} + \sup|J|)$.

Here the values of a field at a point are those on the positive bonds from it, and the Hölder norms are those of its components.

Proof. Throughout, c and $\delta > 0$ denote constants with the dependence stated, which may change from line to line; sums of exponentials $\sum_{y'} e^{-\delta d(y,y')}e^{-\delta' \operatorname{dist}(y',X)}$ over the unit lattice are bounded by a constant times $e^{-\min(\delta,\delta')\operatorname{dist}(y,X)/2}$.

Two local bounds. For a field F and $y \in \mathbb{Z}^4$ let X_y be the set of the points at a distance of at most 9 from y, which contains every bond entering the following estimate: a bond of $\Delta(y)$ lies on staircases from points at a distance of less than 2 from it, these points lie in blocks whose corners are at a distance of less than 2 from them, and the staircases from a block stay within the distance 4 of its corner. First, with $P' := Q_k'^*Q_k'$ acting on each component,

$$\sup_{\Delta(y)}|(P' - Q_k^*Q_k)F| \le 3\|F\|_{X_y}\ . \tag{16.2}$$

Indeed $P'F$ is on $\Delta(y)$ the mean of each component of F over this block, at most $\|F\|_{\Delta(y)}$ in modulus. By the proof of Lemma 6.4(ii), the value of Q_k^*B on a bond of T_η is η times a signed sum of at most $\frac{7}{5\eta}$ values of B, on bonds of the unit lattice from the corners of the blocks of the points whose staircases contain the bond; and by the estimate in the proof of Lemma 9.1(i) each value $(Q_kF)(c)$ is at most $\frac{7}{5}$ times the L^2-norm of F over the bonds of the

staircases from the block of c. Second, by (15.1), (15.2), and the Cauchy–Schwarz inequality on each block,

$$\sup_{\Delta(y)} |\partial P \partial^* F| \le c \sum_{y'} e^{-\delta_\varpi |y-y'|} \|F\|_{\Delta(y')} \,, \tag{16.3}$$

the sum over the unit lattice of the whole space, with F periodic.

The operator G_0. Let G' be the scalar operator $G_{(k)}$ of Section 13 for the constant a, acting on each component of a field. Then $G'^{-1} - G_0^{-1} = a(P' - Q_k^* Q_k)$, hence

$$G_0 = G' + aG'(P' - Q_k^* Q_k) G_0 \,, \tag{16.4}$$

which is [2, (1.133)] with the same constant a in both operators. Let J be bounded, and decompose it into its restrictions $J_{y'}$ to the bonds starting in the blocks $\Delta(y')$, so that $\|J_{y'}\| \le 2 \sup|J|$. By Lemma 16.1 for G_0 and (16.2), the field $h := a(P' - Q_k^* Q_k) G_0 J$ satisfies $\sup_{\Delta(y)} |h| \le c \sum_{y'} e^{-\theta_0 d(y,y')} \sup_{\Delta(y')} |J| \le c\, e^{-\delta \operatorname{dist}(y, \operatorname{supp} J)} \sup|J|$. Applying Proposition 14.3 to J and to the restrictions of h to the blocks, and summing, we obtain the bounds (i) and (ii) for $G_0 J$ and $\nabla G_0 J$. For a tuple J we use (16.4) on $\nabla^* J$: the first term $G' \nabla^* J$ and its derivatives are bounded by Proposition 14.4, and in the second term $h := a(P' - Q_k^* Q_k) G_0 \nabla^* J$ is bounded as before, by the second half of (16.1), so that $G'h$ and $\nabla G'h$ are bounded by Proposition 14.3. This gives (i)–(iv) for G_0 in place of G, except the bound on $\Delta G_0 J$.

The operator G. Since $G_0^{-1} - G^{-1} = \partial P \partial^*$, we have $G = G_0 + G_0 \partial P \partial^* G$, which is [2, (1.132)]. For a bounded field J, Lemma 16.1 for G, applied to the restrictions $J_{y'}$, gives $\|GJ\|_{\Delta(y)} \le c\, e^{-\delta \operatorname{dist}(y, \operatorname{supp} J)} \sup|J|$, and then (16.3) gives the same bound for $\sup_{\Delta(y)} |g|$, $g := \partial P \partial^* G J$. The bounds for G_0, applied to J and to the restrictions of g to the blocks, give (i) and (ii) for GJ and ∇GJ. For a tuple J the same argument with $\nabla^* J$ in place of J, the second half of (16.1), and the bounds for $G_0 \nabla^* J$ and $\nabla G_0 \nabla^* J$ give the bounds for $G \nabla^* J$ and (iii), (iv); in $\nabla G_0 g$ with $g := \partial P \partial^* G \nabla^* J$ only the first derivative of G_0 on a bounded field enters. Finally $\Delta G = I + \partial P \partial^* G - a Q_k^* Q_k G$, and the last two terms applied to J are bounded in the supremum norm by (16.3), by the bound on $Q_k^* Q_k F$ in the proof of (16.2), and by the decay of $\|GJ\|_{\Delta(y)}$. *q.e.d.*

The proposition has the content of [2, (1.110)–(1.113)], with pointwise conditions on the supports in place of the cubes $\tilde{\Delta}(y)$ and the functions ζ there, and with constants which depend on L. With it, Proposition 9.3, and Lemma 16.1, the results of [2] on the operator G hold in the tilted run, except the bounds [2, (1.114)] on the second derivatives in L^2 with their decay, which we have not written out.

17 The Remaining Statements of the First Paper

Three statements of [2] were left aside in Section 10: the bounds on the minimiser H_k, the second derivatives in [2, (1.114)], and the commutators of the averages with functions, from which Bałaban builds his expansions. The second paper on the propagators uses all three, and we settle them here. For the first, Bałaban writes $Q_k G Q_k^*$ in the form [2, (1.99)] and reads its bounds off from the components [2, (1.101)], which are not available for the matrices of the tilted run. We use instead that the inverse of $Q_k G Q_k^*$ is the action Δ_k plus a constant, a fact which follows from the two variational problems defining these operators and does not involve the momentum representation.

The setting is that of Section 16: a torus T_η / Γ', $L \ge 64$, $a > 0$, and $G = \Delta_a^{-1}$. For a positive bond $c = \langle y, y + e_\mu \rangle$ of the unit lattice we write $c_- := y$ and $d(c, c') := d(c_-, c'_-)$, we

let δ_c be the field which is 1 at c and 0 at the other positive bonds, and we put $J_c := Q_k^* \delta_c$. The kernel of an operator M on the fields of the unit lattice is $\mathsf{M}(c,c') := \langle \delta_c, \mathsf{M}\delta_{c'} \rangle$. By (6.3) and the weights of the adjoint, the value of J_c at a positive bond b of T_η is η times the number of the points $x \in B_1(c_-)$ whose staircase $\sigma_\mu^1(x)$ contains b, minus the number of those whose staircase contains b^{-1}. Hence

$$\sup|J_c| \le \tfrac{7}{5} , \qquad \|J_c\|^2 \le 2 , \tag{17.1}$$

by the count in the proof of Lemma 9.1 and by part (i) of that lemma, and J_c vanishes at the bonds which start at a distance of 4 or more from c_-.

Lemma 17.1 (The Operator $Q_k G Q_k^*$)**.** *Let γ_0 and γ_ϕ be the constants of Proposition 9.3.*

(i) Q_k maps the fields on T_η/Γ' onto the fields on the unit lattice, $Q_k G Q_k^*$ is invertible, and $(Q_k G Q_k^*)^{-1} = \Delta_k + a$. The minimiser is $H_k = G Q_k^* (\Delta_k + a)$.

(ii) $(a + 24\gamma_\phi^{-1})^{-1} \le Q_k G Q_k^* \le a^{-1}$ and $Q_k Q_k^* \ge c_Q := \gamma_0 (a + 24\gamma_\phi^{-1})^{-1}$.

(iii) There are constants C and $\theta > 0$, which depend on a and not on k, L, and the torus, such that $|(Q_k G Q_k^*)(c,c')| \le C e^{-\theta d(c,c')}$ and $|(\Delta_k + a)(c,c')| \le C e^{-\theta d(c,c')}$ for all positive bonds c, c' of the unit lattice.

Proof. (i) For every field B on the unit lattice the last part of the proof of Proposition 9.3 exhibits a field A with $Q_k A = B$, so that Q_k is onto and Q_k^* is injective; since G is positive definite, so is $Q_k G Q_k^*$. Fix B and consider the minimum of $\langle A, \Delta_a A \rangle$ over the fields with $Q_k A = B$. It is attained at one field A_0, and $\Delta_a A_0$ is orthogonal to the null space of Q_k, hence of the form $Q_k^* \omega$. Then $A_0 = G Q_k^* \omega$, the constraint gives $\omega = (Q_k G Q_k^*)^{-1} B$, and the minimum is $\langle A_0, \Delta_a A_0 \rangle = \langle \omega, Q_k G Q_k^* \omega \rangle = \langle B, (Q_k G Q_k^*)^{-1} B \rangle$. On the other hand, $\langle A, \Delta_a A \rangle = \|\partial A\|^2 + \|R\partial^* A\|^2 + a\|B\|^2$ on these fields, and by the gauge transformation in the proof of Proposition 9.3 every such field has a companion A_s with $Q_k A_s = B$, $\partial A_s = \partial A$, and $R\partial^* A_s = 0$. The minimum is therefore $a\|B\|^2$ plus the minimum of $\|\partial A\|^2$ over the fields with $Q_k A = B$, which is $\langle B, \Delta_k B \rangle$ by the same proof, and it is attained at a field with $R\partial^* A = 0$, that is, at $H_k B$. Two quadratic forms which agree define the same operator.

(ii) By (9.4), $0 \le \langle B, \Delta_k B \rangle \le \gamma_\phi^{-1} \|\partial_1 B\|^2$, and $\|\partial_1 B\|^2 \le 24 \|B\|^2$, because the value of $\partial_1 B$ at a plaquette is a signed sum of four values of B and a bond lies in six plaquettes. With (i) this gives the bounds on $Q_k G Q_k^*$. By (9.2), $G \le \gamma_0^{-1}$, so that $(a + 24\gamma_\phi^{-1})^{-1} \|\omega\|^2 \le \langle Q_k^* \omega, G Q_k^* \omega \rangle \le \gamma_0^{-1} \|Q_k^* \omega\|^2$.

(iii) We have $(Q_k G Q_k^*)(c,c') = \langle J_c, G J_{c'} \rangle$. A point at a distance of less than 4 from c_- lies in a block $\Delta(y'')$ with $d(y'', c_-) \le 6$, and there are at most 13^4 such blocks. Let $J_{c,y''}$ be the restriction of J_c to the bonds which start in $\Delta(y'')$, so that J_c is the sum of these restrictions and $\|J_{c,y''}\|^2 \le 2$ by (17.1). Lemma 16.1 gives $|\langle J_{c,y''}, G J_{c',y'''} \rangle| \le 2 C_0 e^{-\theta_0 d(y'',y''')}$ with the constant C_0 of that lemma, and $d(y'', y''') \ge d(c,c') - 12$; summing over the pairs of blocks we obtain the first bound, with $\theta = \theta_0$. For the second we apply Lemma 14.1 to $\mathsf{M} := Q_k G Q_k^*$ on the set of the positive bonds of the unit lattice with the function $d(c,c')$, which vanishes for different bonds with the same starting point; the proof of that lemma uses only the triangle inequality. By (ii), $\mathsf{M} \ge \gamma := (a + 24\gamma_\phi^{-1})^{-1}$, and by the first bound $\sum_{c'} |\mathsf{M}(c,c')| (e^{\theta d(c,c')} - 1)$ is at most a constant times $\sum_{y \in \mathbb{Z}^4} e^{-\theta_0 |y|} (e^{\theta |y|} - 1)$, which is at most $\frac{1}{2}\gamma$ for θ small. The lemma and (i) give the second bound. *q.e.d.*

The identity (i) replaces [2, (1.99)], the bounds (ii) are [2, (1.100)] with the conclusion drawn there from [2, (1.101)], and the representation of H_k is [2, (1.103)]. The bounds on H_k now follow from those on G.

Corollary 17.2 (The Minimiser). *For $\alpha \in [0,1)$ there are constants c and $\delta > 0$, which depend on a, L, and α and not on k and the torus, such that for every field B on the unit lattice and all points x, x' of T_η/Γ' with $d(x,x') \le 1$,*

$$\begin{aligned} |(H_k B)(x)| \,,\ |(\nabla H_k B)(x)| &\le c \sum_{c'} e^{-\delta d(x,c'_-)} |B(c')| \,, \\ |(\nabla H_k B)(x) - (\nabla H_k B)(x')| &\le c\, d(x,x')^\alpha \sum_{c'} e^{-\delta d(\{x,x'\},c'_-)} |B(c')| \,. \end{aligned} \tag{17.2}$$

Proof. By Lemma 17.1(i), $H_k B = \sum_c ((\Delta_k + a)B)(c)\, G J_c$. By (17.1) and Proposition 16.2(i) and (ii), GJ_c and $\nabla G J_c$ are at most $\frac{7}{5} c\, e^{4\delta_G} e^{-\delta_G d(x,c_-)}$ in modulus at x, with the corresponding bound for the differences of $\nabla G J_c$, and by Lemma 17.1(iii), $|((\Delta_k + a)B)(c)| \le C \sum_{c'} e^{-\theta d(c,c')} |B(c')|$. The sum over c of the product of the two exponentials is at most a constant times $e^{-\delta d(x,c'_-)}$ with $\delta := \frac{1}{2}\min(\delta_G, \theta)$. *q.e.d.*

We turn to the second derivatives. The decay of Lemma 16.1 was obtained with a weight which is Lipschitz continuous, and such a weight does not commute with the Laplace operator up to a bounded error. The second derivatives are therefore localised afterwards, with a cutoff which has two bounded derivatives. We write $\nabla\nabla A := (\partial'_\nu \partial'_{\nu'} A)_{\nu,\nu'}$, and $\nabla^*\nabla^* J := \sum_{\nu,\nu'} \partial'^*_{\nu'} \partial'^*_\nu J_{\nu\nu'}$ for a family $J = (J_{\nu\nu'})$ of fields.

Lemma 17.3 (Second Derivatives in L^2). *Let $a > 0$. There are constants $\theta_2 > 0$ and C, which depend on a and not on k, L, and the torus, such that for all $y, y' \in \mathbb{Z}^4$ and every field J, or family of fields J, supported in the bonds starting in $\{x \mid d(x,y') \le 2\}$,*

$$\|\nabla\nabla G J\|_{\Delta(y)} \le C e^{-\theta_2 d(y,y')} \|J\| \,, \qquad \|G\nabla^*\nabla^* J\|_{\Delta(y)} \le C e^{-\theta_2 d(y,y')} \|J\| \,. \tag{17.3}$$

Proof. Let $F := GJ$, regarded as a periodic field on T_η, and let $\chi(t) := \prod_\mu \psi(t_\mu - y_\mu - \frac{1}{2})$ for $t \in \mathbb{R}^4$, with a fixed twice continuously differentiable function ψ on $\mathbb{R}$ with values in $[0,1]$ which is 1 on $[-\frac{3}{2}, \frac{3}{2}]$ and 0 outside $[-\frac{5}{2}, \frac{5}{2}]$. Then χF has a finite support, and for a field A of finite support on T_η we have $\|\nabla\nabla A\| = \|\Delta A\|$, because the operators ∂'_ν commute with each other and with their adjoints. The points which enter $\partial'_\nu \partial'_{\nu'} F$ at a point of $\Delta(y)$ have coordinates within 2η of the cube $y + [0,1)^4$, where $\chi = 1$, so that $\|\nabla\nabla F\|_{\Delta(y)} \le \|\Delta(\chi F)\|$. For a point x and each component of F,

$$\begin{aligned} (\Delta(\chi F) - \chi \Delta F)(x) = &-\eta^{-2} \sum_\nu \sum_\pm (\chi(x \pm \eta e'_\nu) - \chi(x))(F(x \pm \eta e'_\nu) - F(x)) \\ &- \eta^{-2} \sum_\nu (\chi(x + \eta e'_\nu) + \chi(x - \eta e'_\nu) - 2\chi(x)) F(x) \,, \end{aligned} \tag{17.4}$$

as one sees by writing out the two second differences. The first differences of χ are at most $\varkappa\eta$ and the second ones at most $\varkappa_2 \eta^2$ in modulus, with constants which depend on ψ only, so that the right side is at most $\varkappa \sum_\nu (|(\partial'_\nu F)(x)| + |(\partial'_\nu F)(x - \eta e'_\nu)|) + 4\varkappa_2 |F(x)|$. The support of χ is contained in the union of the blocks $\Delta(y'')$ with $y'' - y \in \{-2, \dots, 2\}^4$, each of which is mapped injectively to the torus. Hence $\|\Delta(\chi F)\|$ is at most a constant times the sum, over the y'' with $y'' - y \in \{-3, \dots, 3\}^4$, of $\|\Delta F\|_{\Delta(y'')} + \|\nabla F\|_{\Delta(y'')} + \|F\|_{\Delta(y'')}$, the norms being those of the torus.

By Lemma 16.1 the last two norms are at most $C_0 e^{-\theta_0 d(y'',y')} \|J\|$. For the first we use $\Delta F = J + \partial P \partial^* F - a Q_k^* Q_k F$. The norm of J over $\Delta(y'')$ is at most $\|J\|$, and it vanishes unless $d(y'',y') \le 4$. By (16.3) and Lemma 16.1, $\|\partial P \partial^* F\|_{\Delta(y'')} \le$

$2\sup_{\Delta(y'')}|\partial P\partial^* F| \le 2c\sum_{y'''} e^{-\delta_\varpi|y''-y'''|}C_0 e^{-\theta_0 d(y''',y')}\|J\|$, and by the estimates in the proof of (16.2), $\|Q_k^* Q_k F\|_{\Delta(y'')} \le 4\|F\|_{X_{y''}}$, where $X_{y''}$ is contained in the blocks whose corners are at a distance of at most 11 from y''. Since $d(y'',y) \le 6$, all these terms are at most a constant times $e^{-\theta_2 d(y,y')}\|J\|$ with $\theta_2 := \frac{1}{2}\min(\theta_0, \delta_\varpi)$, which is the first bound of (17.3).

For the second, let Φ be a field supported in the bonds starting in $\Delta(y)$. Since G is symmetric, $\langle \Phi, G\nabla^*\nabla^* J\rangle = \langle \nabla\nabla G\Phi, J\rangle$, and the support of J lies in the blocks $\Delta(y'')$ with $d(y'',y') \le 4$. The first bound, with y and y'' exchanged, bounds this by a constant times $e^{-\theta_2 d(y,y')}\|\Phi\|\|J\|$. *q.e.d.*

With Lemma 16.1 this is [2, (1.114)]: for a function ζ supported in $\Delta(y)$ the norms there are at most $\sup|\zeta|$ times the norms over $\Delta(y)$.

It remains to describe the commutators. For a function h on the torus and fields A on T_η/Γ' and B on the unit lattice we write $(hA)(b) := h(b_-)A(b)$ and $(hB)(c) := h(c_-)B(c)$ for positive bonds. Bałaban defines the commutator of Q_k with h through the contours and the segments of his average [2, (1.120)]. The sums of ∂h over paths which occur there are differences of the values of h, so that the definition does not depend on the paths, and it can be stated for any average.

Lemma 17.4 (Commutators with a Function)**.** *Let h be a function on T_η/Γ' with $|h(x) - h(x')| \le \varkappa\, d(x,x')$ for all x, x' and $|h(x+\eta e'_\nu) + h(x-\eta e'_\nu) - 2h(x)| \le \varkappa_2\eta^2$ for all x and ν, and put*

$$S(h) := Q_k h - hQ_k\,, \qquad P_1(h) := h\,\partial P\partial^* - \partial P\partial^* h\,. \tag{17.5}$$

(i) $Q_k^* Q_k h = hQ_k^* Q_k - S(h)^* Q_k + Q_k^* S(h)$, *and* $\Delta_a h = h\Delta_a - K(h)$ *with*

$$K(h) := (h\Delta - \Delta h) + aS(h)^* Q_k - aQ_k^* S(h) - P_1(h)\,. \tag{17.6}$$

(ii) $|(S(h)A)(c)| \le \frac{28}{5}\varkappa\sup|A|$, *the supremum over the bonds which start at a distance of less than 4 from c_-, and $\|S(h)A\| \le 6\varkappa\|A\|$.*

(iii) $P_1(h)$ *has the kernel* $(h(x)-h(x'))\varpi_{\nu\nu'}(x,x')$, *which is at most* $\varkappa c_\varpi |x-x'| e^{-\delta_\varpi|x-x'|}$ *in modulus.*

(iv) $|((h\Delta-\Delta h)A)(x)| \le \varkappa\sum_\nu(|(\partial'_\nu A)(x)| + |(\partial'_\nu A)(x-\eta e'_\nu)|) + 4\varkappa_2|A(x)|$ *for each component of A.*

Proof. (i) By the definition, $Q_k h = hQ_k + S(h)$, and taking adjoints, $hQ_k^* = Q_k^* h + S(h)^*$. Hence $Q_k^* Q_k h = Q_k^* hQ_k + Q_k^* S(h) = hQ_k^* Q_k - S(h)^* Q_k + Q_k^* S(h)$, and the formula for $K(h)$ follows from $\Delta_a = \Delta - \partial P\partial^* + aQ_k^* Q_k$. (ii) By (6.3), $(S(h)A)(c)$ is the average at c of the field $(h - h(c_-))A$. The bonds which enter it start at a distance of less than 4 from c_-, where $|h - h(c_-)| \le 4\varkappa$, and the two bounds follow from the proof of Lemma 9.1(i), in which the average of a field is at most $\frac{7}{5}$ times its supremum, and its square is bounded by a sum over its own bonds. (iii) is (15.1) and (15.2), and (iv) is (17.4) with h in place of χ. *q.e.d.*

Statement (i) is [2, (1.120), (1.121)]. For the functions h_z of [2, (1.118)], which are products of functions of the coordinates x_μ/M_0, the constants are $\varkappa = O(M_0^{-1})$ and $\varkappa_2 = O(M_0^{-2})$, and (ii)–(iv) give the bound [2, (1.128)] on $h_{z_1} K(h_{z_2})$ as in the text. The expansion [2, (1.119)–(1.123)] with its estimates [2, (1.124)–(1.131)] is therefore available in the tilted run, although we have proved Proposition 16.2 without it.

18 What Is Not Covered

The equality (1.1) is not proved. By Proposition 4.2 it is equivalent to a comparison of the construction of the paper on Λ_K, on the torus $\mathbb{T}_\Gamma$, with the tilted run, and for this comparison the following has not been examined.

In [2]: the bounds on H_k through the kernel of $(Q_k G Q_k^*)^{-1}$, the second derivatives in [2, (1.114)], and the commutators of the averages with functions, as described in Section 10.

In [1]: everything which involves an external field $A \neq 0$ or a domain with a boundary. Sections 12–14 treat the whole lattice and the torus at $A = 0$. For a box with Neumann conditions the proof by reflections is not available and no replacement has been carried out; the expansion [1, (2.24)] in the field, the L^p-bounds [1, (2.17)], and Sections 3 to 5 have not been examined.

The other papers on the linear theory. The averaging operations for fields with values in G [4], and [3, 6, 5, 7]. Every background field, propagator, and gauge fixing of the tilted run, at every step, is defined on the turned fine lattice under the constraints of the averages of Section 6.

The coupling. The consequences of the modification stated at the end of Section 5 for the description of the densities after k steps [9, Theorem 1, p. 262] and for the choice of the bare coupling (Paper, Theorem 4.1).

The paper. Its construction on the torus $\mathbb{T}_\Gamma$, whose periods are not multiples of the vectors of the basis; the first step with a loop ensemble (Paper, Theorem 5.3) for the tilted run; and the comparison of two runs, in which the identities of Paper, Lemma 9.2, use of the passage between the two fine lattices only a linear map with a right inverse, while the bounds on the defect (Paper, Proposition 9.3), on the minimisers, and on the large-field terms are proved for two fine lattices of which one is the lattice of the cubes of the other. The normalisation of the tilted run is a product of its own step factors; since ρ fixes a loop in the plane $\{z = 0\}$, the comparison for a single loop relates it to that of the paper whenever the limit for this loop is not zero.

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